

Nonlinear multi-field coupling analysis of piezoelectric semiconductors via PINNs

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Received April 20, 2025; accepted July 8, 2025; published online October 21, 2025

We propose a data-driven physics-informed neural networks (PINNs) via task-decomposition (DD-PINNs-TD) for modeling nonlinear thermal-deformation-polarization-carrier (TDPC) coupling mechanical behaviors of piezoelectric semiconductors (PSs). By embedding three-dimensional (3D), plate, and beam equations of PS structures into the constraints of the DD-PINNs-TD framework, respectively, we develop three representative PINNs that exhibit significant advantages in computational efficiency and accuracy compared to traditional PINNs. Using the proposed DD-PINNs-TD models, we investigate the TDPC coupling responses of PS structures under different loadings. Numerical results demonstrate that the proposed models exhibit accuracy and stability of these models in predicting the nonlinear multi-field coupling mechanical behaviors of PSs. Notably, the plate and beam-theory-based DD-PINNs-TD models achieve superior computational efficiency relative to their 3D-equation-based counterparts. This study establishes a theoretical foundation for analyzing nonlinear multi-field coupling responses in PS structures and has significant practical value in engineering applications.

nonlinear multi-field coupling, piezoelectric semiconductors, structural theories, data-driven, task-decomposition, PINNs

Citation: Z. Xiao, Y. Weng, W. Yao, W. Chen, and C. Zhang, Nonlinear multi-field coupling analysis of piezoelectric semiconductors via PINNs, *Sci. China-Phys. Mech. Astron.* **69**, 214611 (2026), <https://doi.org/10.1007/s11433-025-2742-6>

1 Introduction

Many bulk semiconductors (e.g., ZnO and GaN) [1-3] and two-dimensional (2D) semiconductor materials (e.g., MoS₂ and InSe) [4,5], which consist of one or a few atom layers, possess both semiconductor properties and piezoelectricity. These materials, known as piezoelectric semiconductors (PSs), have attracted enormous attention because of their huge potential in next-generation multifunctional electronic, optoelectronic, and other semiconductor devices [6,7]. The interaction between semiconducting behavior and piezo-

electric polarization effects significantly influences the performance of PS devices. For example, mechanically induced polarization charges near or at the interfaces of Schottky and p-n junctions are utilized to manipulate the gating voltage and depletion layer of these junctions [8,9] and to control quantum phenomena [10]. Theoretical modeling and prediction of the multi-field coupling mechanical behaviors of PS materials and structures are crucial for developing multifunctional PS devices. However, these tasks face significant mathematical challenges due to the nonlinear current equations (reflecting physical nonlinearity) and the possible large deformations (geometrical nonlinearity).

For better understanding the deformation-polarization-

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carrier (DPC) coupling mechanism in PS materials and structures, as well as its effects on PS device performance, researchers have developed linearized methods [11–15] with the assumption of small carrier (i.e., electrons and holes) perturbations to study the influence of the DPC coupling effect on the performance of PS devices in recent years. For example, the DPC fully coupling problems in extension [16,17], bending [18–21], and buckling [22,23] of PS structures under mechanical loads have been analytically studied. Based on the linearized methods, some numerical approaches, including discrete implicit Eulerian finite volume methods (FVMs) [24], characteristic-based mixed finite element methods FEMs [25], meshless local Petrov-Galerkin method [26], and classical FEMs [27,28], have been proposed to investigate the multi-field coupling mechanical behaviors of PS structures. However, the linearized methods are inherently limited in their capacity to treat nonlinear multi-field coupling problems in PS materials and structures.

To address the nonlinear multi-field coupling problems of PSs arising from physical and geometrical nonlinearities, researchers have employed traditional numerical methods to study their nonlinear multi-field coupling responses. For instance, the iterative method has been used to analyze the boundary value problem of PSs [29,30], while a series of finite element schemes employing different types of averaging techniques to discretize 3D continuity equations have been proposed [31]. A stabilized control volume FEM has been developed to provide accurate and stable analysis of drift-diffusion transport in semiconductor devices [32]. Although traditional numerical methods, such as the FEM, have been successful in solving certain nonlinear problems in solid mechanics, they still face several challenges when addressing problems involving multi-field coupling phenomena. These challenges include high computational resource demands [33], difficulties in achieving accurate mesh generation [34], and slow convergence, especially under nonlinear conditions [35].

Recent advancements in machine learning (ML) techniques have provided a powerful alternative approach to solving nonlinear partial differential equations (PDEs), overcoming the abovementioned limitations. For instance, Raissi et al. [36] introduced physics-informed neural networks (PINNs) for solving nonlinear PDEs by embedding the governing physical laws into neural networks. PINNs have been successfully used to address forward and inverse problems of PDEs. For forward problems, comprehensive input data—including governing equations, initial and boundary conditions, and system parameters—are incorporated into the network, enabling efficient handling of complex geometries and high-dimensional problems without the need for meshing. Compared with FEMs, PINNs can offer improved computational efficiency. In inverse problems, PINNs facilitate the optimization and inversion of

physical parameters using known responses, providing effective solutions to challenges associated with incomplete or noisy data. This approach simplifies traditional methods for solving inverse problems by eliminating the need for complete system equations and physical parameters, making it well-suited for data-driven parameter identification challenges.

Recently, PINNs have been successfully extended to the study of solid mechanics problems, including constitutive model identification [37,38], multiscale modeling [39], material parameter estimation and mechanical behavior prediction [40,41], and the prediction of complex multi-field coupling responses [42]. They have been utilized to simulate complex nonlinear behaviors [43], address large deformation issues in elastic materials [44], and predict dynamic responses [45,46]. Furthermore, significant advancements have been made in areas such as design optimization [47–49], crack path prediction [50], and solving corresponding inverse problems [51]. Despite these advances, PINNs still face unresolved issues and limitations [52], including convergence and stability [53], robustness when dealing with complex geometrical structures [54], and challenges in quantifying uncertainties or incomplete quantification in stochastic problems [55].

Particularly, classical PINNs models encounter significant difficulties when applied to complex mechanical problems involving multi-field coupling, heterogeneous materials, and intricate geometrical structures. Challenges include a tendency to converge to local rather than global minima, error accumulation due to significant differences in physical properties, and high computational costs associated with global training over large-scale domains. To address these issues, several domain-decomposition-based PINNs models have been proposed [56–58]. For example, Rezaei et al. [59] introduced a new mixed formulation-based approach combining PINNs and the FEM for heterogeneous domains; Moseley et al. [60] proposed a finite basis PINNs for large-scale and multi-scale problems; Xiang et al. [61] developed a self-adaptive loss balanced PINNs that improves efficiency and accuracy for 2D problems on complex geometrical domains; Dolean et al. [62] introduced multi-level domain-decomposition-based PINNs to enhance efficiency in high-frequency and multi-scale problems; and He et al. [63] proposed a multi-level PINNs for bending analysis of purely elastic structures, with potential applications in digital twin systems. Additionally, conservative PINNs incorporating conservation constraints and domain decomposition techniques have been developed for solving nonlinear conservation laws [64].

As modern computational methods, data-driven techniques bridge the gap between theoretical models and real-world observations, and they hold great potential for solving scientific and engineering problems. The integration of data-

driven techniques with PINNs (DD-PINNs) was pioneered by Raissi et al. [36]. This approach is not only effective for obtaining solutions to PDEs but also for discovering unknown physical laws from various field data. The primary advantages of combining data-driven techniques with PINNs include improved accuracy, robustness to noise and variability, faster training and convergence, and better generalization across complex domains [65]. Following this pioneering work, researchers have successfully predicted the rheological behavior of multi-component complex fluids and the viscosity of nanofluids using low-fidelity data and physical models, outperforming traditional deep neural networks (DNNs) [66,67]. Furthermore, adaptive activation functions integrated with PINNs have proven effective in solving complex nonlinear diffusion-convection-reaction equations [68] and nonlinear local wave problems [69]. Despite these advances, challenges such as uncertainty in unknown parameters, high computational costs, poor convergence, and sensitivity to data quality remain, particularly when addressing multi-field coupling, multi-scale, large-scale, and high-dimensional problems. Moreover, when solving inverse problems, the DD-PINNs method exhibits limitations in model adaptability and generalization capabilities, restricting its application to complex inverse problems involving multi-field coupling and large-scale domains.

To effectively address these challenges in nonlinear multi-field coupling mechanical problems, we propose a novel data-driven PINNs via task-decomposition (DD-PINNs-TD) framework in this paper. It integrates data-driven techniques, task decomposition, and PINNs together. The task decomposition strategy involves dividing the overall problem into multiple subtasks, with each subtask addressing a specific aspect of the problem (e.g., different physical fields or governing equations). These subtasks can be solved either sequentially or concurrently. Unlike existing domain-decomposition-based PINNs methods, the proposed DD-PINNs-TD does not require intricate domain partitioning or complex boundary condition handling. Instead, it divides the overall problem into several subtasks, each managed by an independent neural network. Each network addresses its own boundary conditions and is tailored with specific physical models and network parameters, thereby overcoming limitations of traditional PINNs when handling complex mechanical problems.

By decomposing the global optimization problem into several local optimization problems, the DD-PINNs-TD method significantly enhances prediction accuracy over classical PINNs. The subtask-based local optimization also enables the implementation of parallel computing techniques, further boosting computational efficiency. Importantly, compared with PINNs based on domain decomposition, the proposed method can accurately predict the nonlinear mechanical behavior of structures even in the absence of precise

material parameters by integrating data-driven approaches with PINNs. This eliminates the need for overlapping subdomains and complicated boundary condition management, thereby simplifying the configuration of variable parameters and uncertainties. The enhanced convergence and reduced training time make the DD-PINNs-TD method highly attractive for practical engineering applications, particularly in scenarios involving unknown material parameters and uncertain boundary conditions. Its high reliability and practicality are demonstrated by its ability to accurately capture and predict the mechanical behavior of structures through task decomposition.

The proposed DD-PINNs-TD framework, which integrates data-driven learning with nonlinear multi-field coupled theory, holds significant potential in engineering applications. In particular, it can be applied to the design and optimization of stretchable PS devices, such as wearable sensors, tactile actuators, and electronic skin, where accurate modeling of deformation-polarization-carriers coupling is essential. Moreover, the model enables efficient physical parameter identification and multi-field coupling mechanical response prediction under complex loading conditions, which are critical for real-time performance monitoring, structural health diagnostics, and intelligent control in flexible electronics. These capabilities make the present model a promising tool for accelerating the development of next-generation multifunctional electronic systems.

This paper is organized as follows: Section 2 introduces the overarching concept of the DD-PINNs-TD method and outlines its application to a variety of complex mechanical problems, including multi-field coupling, multi-scale, and multi-level considerations. In Section 3, we establish the DD-PINNs-TD models based on three distinct theoretical frameworks: the 3D theory, plate theory, and beam theory, respectively. Section 4 employs the DD-PINNs-TD method to predict the nonlinear multi-field coupling mechanical behaviors of the 3D PS structure, plate-like structure, and beam-like structure under different loading conditions, respectively, and contrasts these results with the COMSOL model. In addition, a comparison with traditional PINNs is conducted to demonstrate the advantages of the proposed method in terms of accuracy and computational time cost. Finally, Section 5 encapsulates our conclusions.

2 Construction of data-driven PINNs via task-decomposition

To gain a better understanding of nonlinear multi-field coupling mechanical problems, particularly those involving the prediction and description of physical laws, we construct the model of the DD-PINNs-TD method in this section. This model is specifically tailored for the third-/fourth-generation

semiconductors that possess deformation-polarization-carriers coupling characteristics. Figure 1(a) shows the process of solving solid mechanics problems using the DD-PINNs-TD method. Generally, the overall mechanical problem is treated as a primary task that can be divided into multiple first-level subtasks (e.g., first-level subtasks: FLS-1, FLS-2, ..., FLS- M), as depicted in Figure 1(b), (c), and (d). In turn, each first-level subtask may be divided into second-level subtasks (e.g., second-level subtasks: SLS-1, SLS-2, ..., SLS- M). In this hierarchical approach, the complete mechanical problem is regarded as a composite task, with each sub-problem constituting a component of the overall problem.

The neural network architecture within our framework can be designed with flexibility in mind, allowing it to be adapted to various mechanical problems. For example, (1) multi-field coupling problems: as illustrated in Figure 1(b), when addressing problems involving multiple physical fields, the overall problem can be segmented into distinct sub-problems corresponding to individual fields (such as the elastic field, electromechanical field, and magnetoelectric field). Similarly, the physical domain itself can be divided into multiple subdomains, with each domain or field treated as a separate subtask (e.g., SLS-1 and SLS-2); (2) multilevel structural problems: in cases involving multilevel structures, as depicted in Figure 1(c), each structural level is considered a distinct subtask (e.g., FLS-1 for the first-level structure and FLS-2 for the second-level structure). If a particular level comprises various cell structures, a secondary task decomposition (resulting in subtasks like SLS-1, SLS-2) is performed to capture the detailed behavior at that level; and (3) multiscale problems: for problems spanning multiple

scales, see Figure 1(d), the subproblems may be organized to cover different scales, including the microscale, mesoscale, and macroscale, thereby enabling a comprehensive analysis of the system.

In this paper, we illustrate the application of the DD-PINNs-TD framework by addressing a nonlinear multi-field coupling problem in phase-separation (PS) systems. Specifically, we demonstrate the effectiveness of our volume segmentation method by accurately predicting the nonlinear multi-field mechanical behavior in both beam- and plate-like PS structures. To support the training process, a dedicated dataset is constructed for each subtask or field. The datasets are organized with corresponding labels, for example, training dataset 1 for FLS-1, training dataset 2 for FLS-2, and so forth up to training dataset M for FLS- M . With this approach, the construction and organization of the training data are fully completed, setting the stage for efficient model training and subsequent validation.

Subsequently, the corresponding neural networks and physics-informed components are constructed based on the different training datasets, as depicted in Figure 2. Referring back to Figure 1, we recall the process of partitioning the overall mechanical problem and creating the relevant training datasets. Assuming there are M subtasks (as shown in Figure 1), this results in the establishment of M distinct neural network architectures, as illustrated in Figure 2. Each neural network architecture can be configured with various parameters, such as the number of hidden layers, neurons per layer, and the type of activation function.

For each subtask within the DD-PINNs-TD model, it is necessary to construct the corresponding physical model, essentially a learning function (f_a^k), to improve prediction

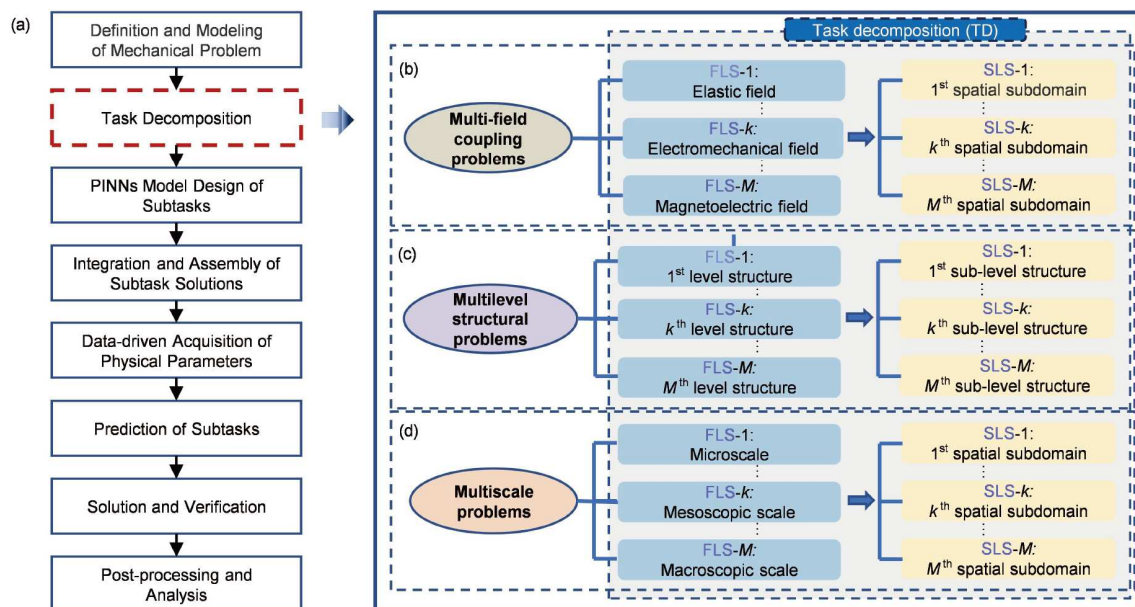


Figure 1 (Color online) (a) The process of solving mechanics problems by the DD-PINNs-TD method and the composition of various mechanical problems based on the DD-PINNs-TD method; (b) multi-field coupling problems; (c) multilevel structural problems; (d) multiscale problems.

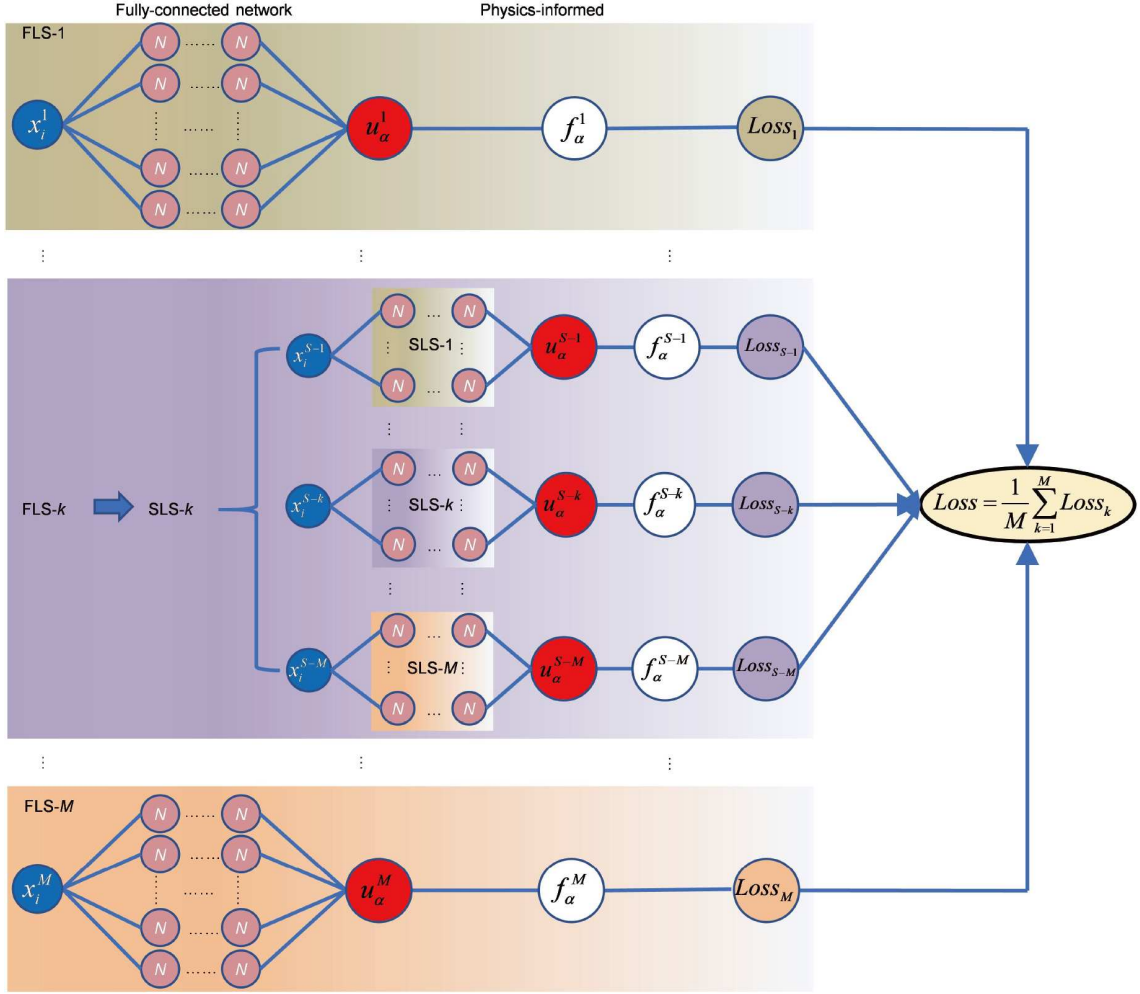


Figure 2 (Color online) The neural network architecture of the DD-PINNs-TD method.

accuracy. If there exists any coupling or shared information between the sub-physical problems or among different regions, continuity conditions can be incorporated into the physical information. For the k th neural network architecture, the loss function is defined as follows:

$$Loss_k = L_u^k + L_f^k, \quad (1)$$

where L_u^k and L_f^k , considering the error terms corresponding to the physical variables and the intrinsic physical laws governing the studied problem, are given as:

$$L_u^k = \frac{1}{N_k} \sum_{l=1}^{N_k} \sum_{j=1}^{\alpha} |u_j^{\text{Pre}}(x_l) - u_j^{\text{Ref}}(x_l)|^2, \quad (2)$$

$$L_f^k = \frac{1}{N_k} \sum_{l=1}^{N_k} \sum_{j=1}^{\alpha} |f_j^k(x_l)|^2, \quad (3)$$

where $u_j^{\text{Pre}}(x_l)$ and $f_j^k(x_l)$ represent the predicted physical quantities and the learning function evaluated at a set of collocation points, while $u_j^{\text{Ref}}(x_l)$ denotes the training data available at those same points. Then, the loss function for the DD-PINNs-TD model can be given by aggregating the losses

from each subtask:

$$Loss = \frac{1}{M} \sum_{k=1}^M Loss_k. \quad (4)$$

In practice, the original data, whether obtained from simulation data or experimental data, must be normalized to ensure that the loss terms are on the same order of magnitude. In this study, a linear normalization formula is used as:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (5)$$

where x represents the original data, $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of the original dataset, respectively. For further details regarding the implementation process, please refer to Appendix A.

3 DD-PINNs-TD model for nonlinear multi-field coupling mechanics of PS structures

In this section, we develop the DD-PINNs-TD model to predict the nonlinear multi-field coupling mechanical beha-

vior of n-type PS structures by embedding the governing equations, which describe the underlying physical laws of PS structures, directly into the network configuration. This approach follows the general framework of the DD-PINNs-TD model described in the previous section.

For a general 3D PS structure with dimensions a (length) $\times b$ (width) $\times c$ (height) and an initial electron concentration (N_0), as shown in Figure 3(a), the corresponding 3D governing equations are incorporated into the neural network. In practice applications, especially in semiconductor devices, beam-like and plate-like structures are often used as fundamental building blocks, as illustrated in Figure 3(b) and (c). Although the 3D theory-based DD-PINNs-TD model can be applied to these cases, we opt to integrate the governing equations from beam theory and plate theory into the neural network for these specific geometries. These two specialized models—referred to as the beam theory-based model and the plate theory-based model (denoted as beam theory-based DD-PINNs-TD and plate theory-based DD-PINNs-TD)—exploit the advantages of structural theories to strike an effective balance between computational efficiency and accuracy. These three types of DD-PINNs-TD models lay a solid foundation for modeling multi-field coupling mechanical problems of PS structures and their corresponding devices from a continuum perspective.

3.1 3D theory-based DD-PINNs-TD model

To establish the 3D theory-based DD-PINNs-TD framework, we begin by summarizing the 3D nonlinear basic equations for PS structures. For the n-type PS structure, in the absence of any body forces, the governing equations, including the static equilibrium equations, the charge equation of electrostatics (Gauss's law), the continuity equation for electrons, and the conservation equation of heat, are given as:

$$\begin{aligned} T_{ij,j} &= 0, \\ D_{i,i} &= q(-n + N_D^+), \\ J_{i,i} &= 0, \\ h_{i,i} &= J_i \cdot E_i, \end{aligned} \quad (6)$$

where T_{ij} , D_i , J_i , n , h_i , and E_i represent the stress, electric

displacement, electron current density, electron concentration, heat flux, and electric field components, respectively, and q the elementary charge. In eq. (6), only Joule heating is taken into consideration. However, it is straightforward to incorporate additional heat sources, such as those arising from thermoelectric effects, by extending the formulation accordingly.

The constitutive equations of stress, electric displacement, electron current density, and heat flux components are written in the form of strain (S_{kl}), electric field (E_k), temperature change (θ), and electron concentration (n) as:

$$\begin{aligned} T_{ij} &= c_{ijkl} S_{kl} - e_{kij} E_k - g_{ij} \theta, \\ D_k &= e_{kij} S_{ij} + \epsilon_{kl} E_k + \bar{p}_k \theta, \\ J_k &= qn\mu_{ij} E_k + qD_{ij} n_{,k}, \\ h_i &= -\kappa_{ij} \theta_{,j}, \end{aligned} \quad (7)$$

where c_{ijkl} , e_{ij} , g_{ij} , ϵ_{kl} , μ_{ij} , D_{ij} , and κ_{ij} are the elastic modulus, piezoelectric constant, thermal stress coefficient, dielectric constant, electron mobility, electron diffusion coefficient, and thermal conductivity, respectively.

The nonlinear relationship between strain (S_{ij}) and mechanical displacement (u_i) and the gradient relation between electric potential (φ) and electric field are described by

$$S_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i} + u_{k,i} u_{k,j}), \quad E_i = -\varphi_{,i}. \quad (8)$$

In addition, we denote the gradients of electron concentration (n) and temperature change (θ) as:

$$\bar{n}_i = n_{,i}, \quad \Theta_i = -\theta_{,i}. \quad (9)$$

By substituting eqs. (7)–(9) into the governing equations given in eq. (6) and recasting them into matrix form, we have

$$\mathbf{K}_\beta \mathbf{u}_\beta = \mathbf{q}, \quad (10)$$

where $\mathbf{K}_\beta$ ($\beta=1,2,3,\dots,25$), $\mathbf{u}_\beta$, and $\mathbf{q}$ represent the parameter matrix, the physical quantities matrix, and the load matrix, respectively. The details of these matrixes ($\mathbf{K}_\beta$, $\mathbf{u}_\beta$, $\mathbf{q}$) are provided in Appendix B.

To incorporate the physical laws described by eq. (10) into the PINNs framework, we need to convert eq. (10) into a general form suitable for PINNs:

$$u_t + N[u; \lambda] = 0, \quad x \in \Omega, \quad t \in [0, T]. \quad (11)$$

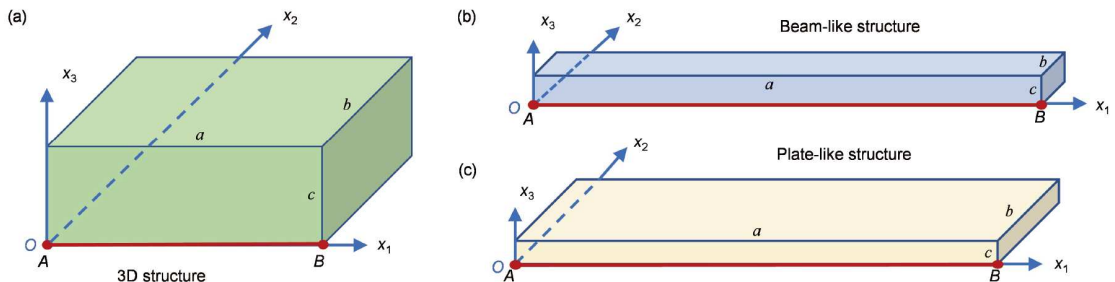


Figure 3 (Color online) Sketch of PS structures: (a) 3D structure; (b) beam-like structure; (c) plate-like structure.

Thus, eq. (10) can be rewritten as:

$$\mathbf{N}_\beta \mathbf{u}_\beta = \mathbf{q}, \quad (12)$$

where $\mathbf{N}_\beta$ is a nonlinear operator parametrized by an unknown parameter λ_β^{3D} . Details of the nonlinear operator ($\mathbf{N}_\beta$) are given in eq. (B.4) in Appendix B.

Following the solution process of the DD-PINNs-TD model illustrated in Figure 2(b), we define the learning function for the k th subtask as:

$$f_\beta^k(\mathbf{x}) = \mathbf{N}_\beta \mathbf{u}_\beta - \mathbf{q}, \quad (13)$$

where $\mathbf{x}=(x_1, x_2, x_3)$ represents the space coordinate point. We then define the mean square error (MSE) between the predicted and reference values of the physical quantity matrix $\mathbf{u}$, see eq. (1). The specific formulation of this loss function is detailed in Appendix B, see eq. (B.5).

To ensure smooth transitions between the adjacent subtasks, the points on the boundaries between these subtasks must satisfy the conditions:

$$\begin{aligned} \bar{u}_i^k &= (u_i^k + u_i^{k-1}) / 2, & \bar{\varphi}^k &= (\varphi^k + \varphi^{k-1}) / 2, \\ \bar{n}^k &= (n^k + n^{k-1}) / 2. \end{aligned} \quad (14)$$

Figure A.1 shows the 3D theory-based DD-PINNs-TD network architecture used for solving nonlinear multi-field coupling problems of PS structures. It confirms that the solution process of the DD-PINNs-TD model is consistent with the schematic provided in Figure 1(a). In practice, the original dataset, including boundary condition data and result data, is first normalized using the same normalization formula as given in eq. (5) to create the training dataset. This normalized dataset is then input into the fully-linked neural network model, where the physical information constraints are imposed according to eq. (12). The L-BFGS optimizer is subsequently used to minimize the loss function during training. When the loss function value, denoted as $Loss$, is minimized, the model has completed its training, and the value of the parameter λ_β^{3D} is obtained. This value is then incorporated into the physical information constraint neural network for the predictive analysis.

3.2 Beam theory-based DD-PINNs-TD model

Since beam-like structures consist of important building blocks in various engineering applications. Hence, we develop the beam theory-based DD-PINNs-TD model in this subsection. Then, based on eqs. (C.17)-(C.25) derived in Appendix C, we can develop the beam theory-based DD-PINNs-TD model. In doing so, we introduce parameters λ_β^B ($\beta=1, 2, \dots, 17$) to replace the corresponding elastic, piezoelectric, dielectric, thermal stress, electron mobility, and diffusion constants. With these substitutions, eqs. (C.17)-

(C.25) can be rewritten as:

$$\lambda_1^B u_{1,11}^{(0)} + \lambda_1^B u_{3,11}^{(0)} u_{3,11}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)} = 0, \quad (15)$$

$$\lambda_3^B u_{1,11}^{(1)} + \lambda_3^B u_1^{(1)} + \lambda_3^B u_{3,1}^{(0)} + \lambda_4^B \varphi_{,1}^{(0)} = 0, \quad (16)$$

$$\begin{aligned} & \left(\lambda_1^B u_{1,11}^{(0)} u_{3,1}^{(0)} + \lambda_1^B u_{3,1}^{(0)} u_{3,11}^{(0)} u_{3,1}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)} u_{3,1}^{(0)} \right) \\ & + \left(\lambda_1^B u_{1,11}^{(0)} u_{3,11}^{(0)} + 0.5 \lambda_1^B \left(u_{3,1}^{(0)} \right)^2 u_{3,11}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)} u_{3,11}^{(0)} \right) \\ & + \left(\lambda_3^B u_{1,11}^{(1)} + \lambda_3^B u_{3,11}^{(0)} + \lambda_4^B \varphi_{,1}^{(0)} \right) = 0, \end{aligned} \quad (17)$$

$$\lambda_4^B u_{1,11}^{(1)} + \lambda_4^B u_{3,11}^{(0)} - \lambda_{10}^B \varphi_{,11}^{(0)} + \lambda_{17}^B (n^{(0)} - N_0) = 0, \quad (18)$$

$$\begin{aligned} & -\lambda_7^B \varphi_{,11}^{(1)} + \lambda_2^B u_{1,11}^{(0)} + 0.5 \lambda_2^B \left(u_{3,1}^{(0)} \right)^2 \\ & -\lambda_{11}^B \varphi_{,11}^{(1)} + \lambda_{12}^B n^{(1)} = 0, \end{aligned} \quad (19)$$

$$\begin{aligned} & -\lambda_5^B n_{,1}^{(0)} \varphi_{,1}^{(0)} - \lambda_5^B n^{(0)} \varphi_{,11}^{(0)} - \lambda_6^B n_{,1}^{(1)} \varphi_{,1}^{(1)} \\ & -\lambda_6^B n^{(1)} \varphi_{,11}^{(1)} + \lambda_8^B n_{,11}^{(0)} = 0, \end{aligned} \quad (20)$$

$$\begin{aligned} & -\lambda_6^B n_{,1}^{(0)} \varphi_{,1}^{(1)} - \lambda_6^B n^{(0)} \varphi_{,11}^{(1)} - \lambda_6^B n_{,1}^{(1)} \varphi_{,1}^{(0)} - \lambda_6^B n^{(1)} \varphi_{,11}^{(0)} \\ & + \lambda_9^B n_{,11}^{(1)} - \lambda_5^B n^{(0)} \varphi_{,1}^{(1)} + \lambda_8^B n^{(1)} = 0, \end{aligned} \quad (21)$$

$$\lambda_{14}^B \Theta_{1,1}^{(0)} + \lambda_{15}^B \Theta_{1,1}^{(1)} = s^{(0)}, \quad (22)$$

$$\lambda_{15}^B \Theta_{1,1}^{(0)} + \lambda_{16}^B \Theta_{1,1}^{(1)} + \lambda_{14}^B \Theta_3^{(0)} = 0. \quad (23)$$

We assume that the overall task is decomposed into M subtasks, with the corresponding neural network architecture divided into M subnetworks accordingly. If the physical information for each subtask in the DD-PINNs-TD model is identical, then for the k th subtask, we define the learning function $f_i^k(x_1)$ as:

$$\begin{aligned} f_1^k &= \lambda_1^B u_{1,11}^{(0)} + \lambda_1^B u_{3,11}^{(0)} u_{3,11}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)}, \\ f_2^k &= \lambda_3^B u_{1,11}^{(1)} + \lambda_3^B u_1^{(1)} + \lambda_3^B u_{3,1}^{(0)} + \lambda_4^B \varphi_{,1}^{(0)}, \\ f_3^k &= \left(\lambda_1^B u_{1,11}^{(0)} u_{3,1}^{(0)} + \lambda_1^B u_{3,1}^{(0)} u_{3,11}^{(0)} u_{3,1}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)} u_{3,1}^{(0)} \right) \\ & + \left(\lambda_1^B u_{1,11}^{(0)} u_{3,11}^{(0)} + 0.5 \lambda_1^B \left(u_{3,1}^{(0)} \right)^2 u_{3,11}^{(0)} + \lambda_2^B \varphi_{,1}^{(1)} u_{3,11}^{(0)} \right) \\ & + \left(\lambda_3^B u_{1,11}^{(1)} + \lambda_3^B u_{3,11}^{(0)} + \lambda_4^B \varphi_{,1}^{(0)} \right), \\ f_4^k &= \lambda_4^B u_{1,11}^{(1)} + \lambda_4^B u_{3,11}^{(0)} - \lambda_{10}^B \varphi_{,11}^{(0)} + \lambda_{17}^B (n^{(0)} - N_0), \\ f_5^k &= -\lambda_7^B \varphi_{,11}^{(1)} + \lambda_2^B u_{1,11}^{(0)} + 0.5 \lambda_2^B \left(u_{3,1}^{(0)} \right)^2 \\ & -\lambda_{11}^B \varphi_{,11}^{(1)} + \lambda_{12}^B n^{(1)}, \\ f_6^k &= -\lambda_5^B n_{,1}^{(0)} \varphi_{,1}^{(0)} - \lambda_5^B n^{(0)} \varphi_{,11}^{(0)} - \lambda_6^B n_{,1}^{(1)} \varphi_{,1}^{(1)} \\ & -\lambda_6^B n^{(1)} \varphi_{,11}^{(1)} + \lambda_8^B n_{,11}^{(0)}, \\ f_7^k &= -\lambda_6^B n_{,1}^{(0)} \varphi_{,1}^{(1)} - \lambda_6^B n^{(0)} \varphi_{,11}^{(1)} - \lambda_6^B n_{,1}^{(1)} \varphi_{,1}^{(0)} \\ & -\lambda_6^B n^{(1)} \varphi_{,11}^{(0)} + \lambda_9^B n_{,11}^{(1)} - \lambda_5^B n^{(0)} \varphi_{,1}^{(1)} \\ & + \lambda_8^B n^{(1)}, \\ f_8^k &= \lambda_{14}^B \Theta_{1,1}^{(0)} + \lambda_{15}^B \Theta_{1,1}^{(1)} - s^{(0)}, \\ f_9^k &= \lambda_{15}^B \Theta_{1,1}^{(0)} + \lambda_{16}^B \Theta_{1,1}^{(1)} + \lambda_{14}^B \Theta_3^{(0)}. \end{aligned} \quad (24)$$

If the physical information differs among subtasks, the appropriate physical model is embedded within the corresponding neural network branch. In our implementation, we assume the same physical model is used for all subtasks in solving the nonlinear multi-field coupling problem of beam-like PS structures. The prediction accuracy is assessed using a loss function. For the k th subtask, the loss function is defined similarly to eq. (1), see eq. (B.7) given in Appendix B for details. To ensure smooth transitions between subtasks, the points at the boundaries of adjacent subtasks satisfy the condition described in eq. (14).

The overall beam theory-based DD-PINNs-TD network architecture for solving the nonlinear multi-field coupling problem of beam-like PS structures is shown in Appendix A, see Figure A.3. The training and prediction process using the beam theory-based DD-PINNs-TD model are illustrated in Figure 2(b).

3.3 Plate theory-based DD-PINNs-TD model

Similar to beam-like structures, plate-like structures also serve as essential building blocks in a wide range of engineering applications. In this subsection, we develop a plate theory-based DD-PINNs-TD model by integrating the governing physical laws that describe the multi-field coupling mechanical behavior of plate-like PS structures. The basic equations of the plate theory for PS structures have been presented in ref. [70] (for brevity, these equations are not reproduced here). To simplify the representation of material properties within the plate theory-based model, we introduce a set of parameters λ_β^P ($\beta=1, 2, \dots, 30$) to replace the corresponding elastic, piezoelectric, dielectric, thermal stress, mobility, and diffusion constants. With these substitutions, the governing equations originally given in eqs. (A.1)–(A.11) in ref. [70] can be rewritten as:

$$\lambda_1^P(u_{1,11}^{(0)} + u_{3,1}^{(0)}u_{3,11}^{(0)}) + \lambda_2^P(u_{2,21}^{(0)} + u_{3,2}^{(0)}u_{3,21}^{(0)}) + \lambda_3^P\varphi_{,1}^{(1)} - \lambda_4^P\theta_{,1}^{(0)} + \lambda_5^P(u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)}u_{3,2}^{(0)})_{,2} = 0, \quad (25)$$

$$\lambda_5^P(u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)}u_{3,2}^{(0)})_{,1} + \lambda_6^P(u_{1,12}^{(0)} + u_{3,1}^{(0)}u_{3,12}^{(0)}) + \lambda_7^P(u_{2,22}^{(0)} + u_{3,2}^{(0)}u_{3,22}^{(0)}) + \lambda_8^P\varphi_{,2}^{(1)} - \lambda_4^P\theta_{,2}^{(0)} = 0, \quad (26)$$

$$\lambda_1^P\left[u_{1,1}^{(0)} + \frac{1}{2}u_{3,1}^{(0)}u_{3,1}^{(0)}\right]u_{3,1}^{(0)} + \lambda_2^P\left[u_{2,2}^{(0)} + \frac{1}{2}u_{3,2}^{(0)}u_{3,2}^{(0)}\right]u_{3,2}^{(0)} + \lambda_3^P(\varphi_{,1}^{(1)}u_{3,1}^{(0)})_{,1} - \lambda_4^P[(\theta_{,1}^{(0)}u_{3,1}^{(0)})_{,1} + (\theta_{,2}^{(0)}u_{3,2}^{(0)})_{,2}] + \lambda_5^P[(u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)}u_{3,2}^{(0)})u_{3,2}^{(0)}]_{,1}$$

$$+ \lambda_5^P\left[u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)}u_{3,2}^{(0)}\right]u_{3,1}^{(0)} + \lambda_6^P\left[u_{1,1}^{(0)} + \frac{1}{2}u_{3,1}^{(0)}u_{3,1}^{(0)}\right]u_{3,2}^{(0)} + \lambda_7^P\left[u_{2,2}^{(0)} + \frac{1}{2}u_{3,2}^{(0)}u_{3,2}^{(0)}\right]u_{3,2}^{(0)} + \lambda_8^P(\varphi_{,1}^{(1)}u_{3,2}^{(0)})_{,2} + \lambda_9^P(u_{3,1}^{(0)} + u_1^{(1)})_{,1} + \lambda_9^P(u_{3,2}^{(0)} + u_2^{(1)})_{,2} + \lambda_{10}^P(\varphi_{,11}^{(0)} + \varphi_{,22}^{(0)}) = 0, \quad (27)$$

$$\lambda_{11}^Pu_{1,11}^{(1)} + \lambda_{12}^Pu_{2,21}^{(1)} - \lambda_{13}^P\theta_{,1}^{(1)} + \lambda_{14}^P(u_{1,2}^{(1)} + u_{2,1}^{(1)})_{,2} - \lambda_9^P(u_{3,1}^{(0)} + u_1^{(1)}) - \lambda_{10}^P\varphi_{,1}^{(0)} = 0, \quad (28)$$

$$\lambda_{14}^P(u_{1,1}^{(1)} + u_{2,2}^{(1)})_{,2} + \lambda_{15}^Pu_{1,12}^{(1)} + \lambda_{16}^Pu_{2,22}^{(1)} - \lambda_{13}^P\theta_{,2}^{(1)} - \lambda_9^P(u_{3,2}^{(0)} + u_2^{(1)}) - \lambda_{17}^P\varphi_{,2}^{(0)} = 0, \quad (29)$$

$$\lambda_{18}^P[(u_{3,1}^{(0)} + u_1^{(1)})_{,1} + (u_{3,2}^{(0)} + u_2^{(1)})_{,2}] - \lambda_{19}^P(\varphi_{,11}^{(0)} + \varphi_{,22}^{(0)}) + \lambda_{20}^P(\theta_{,1}^{(0)} + \theta_{,2}^{(0)}) + \lambda_{30}^P(n^{(0)} - N_0) = 0, \quad (30)$$

$$- \lambda_{21}^P(\varphi_{,11}^{(1)} + \varphi_{,22}^{(1)}) + \lambda_{23}^P\varphi^{(1)} + \lambda_{22}^P(\theta_{,1}^{(1)} + \theta_{,2}^{(1)}) - \lambda_{20}^P\theta^{(0)} - \lambda_3^P(u_{1,1}^{(0)} + 0.5u_{3,1}^{(0)}u_{3,1}^{(0)}) - \lambda_8^P(u_{2,2}^{(0)} + 0.5u_{3,2}^{(0)}u_{3,2}^{(0)}) = 0, \quad (31)$$

$$- \lambda_{24}^P[(n^{(0)}\varphi_{,1}^{(0)})_{,1} + (n^{(0)}\varphi_{,2}^{(0)})_{,2}] - \lambda_{25}^P[(n^{(1)}\varphi_{,1}^{(1)})_{,1} + (n^{(1)}\varphi_{,2}^{(1)})_{,2}] + \lambda_{26}^P(n_{,11}^{(0)} + n_{,22}^{(0)}) = 0, \quad (32)$$

$$- \lambda_{25}^P[(n^{(0)}\varphi_{,1}^{(1)} + n^{(1)}\varphi_{,1}^{(0)})_{,1} + (n^{(0)}\varphi_{,2}^{(1)} + n^{(1)}\varphi_{,2}^{(0)})_{,2}] + \lambda_{27}^P(n_{,11}^{(1)} + n_{,22}^{(1)}) - \lambda_{24}^Pn^{(1)}\varphi^{(1)} + \lambda_{26}^Pn^{(1)} = 0, \quad (33)$$

$$\lambda_{28}^P(\theta_{,11}^{(0)} + \theta_{,22}^{(0)}) + s^{(0)} = 0, \quad (34)$$

$$\lambda_{29}^P(\theta_{,11}^{(1)} + \theta_{,22}^{(1)}) - \lambda_{28}^P\theta^{(1)} = 0. \quad (35)$$

Similarly, suppose that the overall problem is decomposed into M subtasks and the corresponding neural network architecture is partitioned into M subnetworks. Furthermore, assume that the physical information for each subtask in the DD-PINNs-TD model is identical. Then, for the k th subtask, we can define the corresponding learning function $f_i^k(x_1, x_2)$ as:

$$f_1^k = \lambda_1^Pu_{1,11}^{(0)} + \lambda_1^Pu_{3,1}^{(0)}u_{3,11}^{(0)} + \lambda_2^Pu_{2,21}^{(0)} + \lambda_2^Pu_{3,2}^{(0)}u_{3,21}^{(0)} + \lambda_3^P\varphi_{,1}^{(1)} - \lambda_4^P\theta_{,1}^{(0)} + \lambda_5^Pu_{1,22}^{(0)} + \lambda_5^Pu_{2,12}^{(0)} + \lambda_5^Pu_{3,12}^{(0)}u_{3,2}^{(0)} + \lambda_5^Pu_{3,1}^{(0)}u_{3,22}^{(0)},$$

$$f_2^k = \lambda_5^P(u_{1,21}^{(0)} + u_{2,11}^{(0)} + u_{3,1}^{(0)}u_{3,2}^{(0)} + u_{3,1}^{(0)}u_{3,21}^{(0)}) + \lambda_6^P(u_{1,12}^{(0)} + u_{3,1}^{(0)}u_{3,12}^{(0)}) + \lambda_7^P(u_{2,22}^{(0)} + u_{3,2}^{(0)}u_{3,22}^{(0)}) + \lambda_8^P\varphi_{,2}^{(1)} - \lambda_4^P\theta_{,2}^{(0)},$$

$$\begin{aligned}
f_3^k &= \lambda_9^P (u_{3,11}^{(0)} + u_{1,1}^{(1)}) + \lambda_{10}^P \varphi_{,11}^{(0)} + \lambda_9^P (u_{3,22}^{(0)} + u_{2,2}^{(1)}) \\
&\quad + \lambda_{10}^P \varphi_{,22}^{(0)} + \lambda_1^P (u_{1,11}^{(0)} + u_{3,1}^{(0)} u_{3,11}^{(0)}) u_{3,1}^{(0)} \\
&\quad + \lambda_1^P (u_{1,1}^{(0)} + 0.5 u_{3,1}^{(0)} u_{3,11}^{(0)}) u_{3,11}^{(0)} \\
&\quad + \lambda_2^P (u_{2,21}^{(0)} + u_{3,2}^{(0)} u_{3,21}^{(0)}) u_{3,1}^{(0)} \\
&\quad + \lambda_2^P (u_{2,2}^{(0)} + 0.5 u_{3,2}^{(0)} u_{3,21}^{(0)}) u_{3,11}^{(0)} \\
&\quad + \lambda_3^P \varphi_{,1}^{(1)} u_{3,1}^{(0)} + \lambda_3^P \varphi_{,11}^{(1)} u_{3,11}^{(0)} - \lambda_4^P \theta_{,1}^{(0)} u_{3,1}^{(0)} \\
&\quad - \lambda_4^P \theta_{,11}^{(0)} u_{3,11}^{(0)} \\
&\quad + \lambda_5^P (u_{1,21}^{(0)} + u_{2,11}^{(0)} + u_{3,11}^{(0)} u_{3,2}^{(0)} + u_{3,1}^{(0)} u_{3,21}^{(0)}) u_{3,2}^{(0)} \\
&\quad + \lambda_5^P (u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)} u_{3,2}^{(0)}) u_{3,21}^{(0)} \\
&\quad + \lambda_6^P (u_{1,12}^{(0)} + u_{3,1}^{(0)} u_{3,12}^{(0)}) u_{3,2}^{(0)} \\
&\quad + \lambda_6^P (u_{1,1}^{(0)} + 0.5 u_{3,1}^{(0)} u_{3,12}^{(0)}) u_{3,22}^{(0)} \\
&\quad + \lambda_7^P (u_{2,22}^{(0)} + u_{3,2}^{(0)} u_{3,22}^{(0)}) u_{3,2}^{(0)} \\
&\quad + \lambda_7^P (u_{2,2}^{(0)} + 0.5 u_{3,2}^{(0)} u_{3,22}^{(0)}) u_{3,22}^{(0)} \\
&\quad + \lambda_8^P \varphi_{,2}^{(1)} u_{3,2}^{(0)} + \lambda_8^P \varphi_{,22}^{(1)} u_{3,22}^{(0)} - \lambda_4^P \theta_{,2}^{(0)} u_{3,2}^{(0)} \\
&\quad - \lambda_4^P \theta_{,22}^{(0)} u_{3,22}^{(0)} \\
&\quad + \lambda_5^P (u_{1,22}^{(0)} + u_{2,12}^{(0)} + u_{3,12}^{(0)} u_{3,2}^{(0)} + u_{3,1}^{(0)} u_{3,22}^{(0)}) u_{3,1}^{(0)} \\
&\quad + \lambda_5^P (u_{1,2}^{(0)} + u_{2,1}^{(0)} + u_{3,1}^{(0)} u_{3,2}^{(0)}) u_{3,12}^{(0)} \\
f_4^k &= \lambda_{11}^P u_{1,11}^{(1)} + \lambda_{12}^P u_{2,21}^{(1)} - \lambda_{13}^P \theta_{,1}^{(1)} + \lambda_{14}^P (u_{1,22}^{(1)} + u_{2,12}^{(1)}) \\
&\quad - \lambda_9^P (u_{3,1}^{(0)} + u_{1,1}^{(1)}) - \lambda_{10}^P \varphi_{,1}^{(0)}, \\
f_5^k &= \lambda_{14}^P (u_{1,21}^{(1)} + u_{2,22}^{(1)}) + \lambda_{15}^P u_{1,12}^{(1)} + \lambda_{16}^P u_{2,22}^{(1)} - \lambda_{13}^P \theta_{,2}^{(1)} \\
&\quad - \lambda_9^P (u_{3,2}^{(0)} + u_{2,2}^{(1)}) - \lambda_{17}^P \varphi_{,2}^{(0)}, \\
f_6^k &= \lambda_{18}^P (u_{3,11}^{(0)} + u_{1,1}^{(1)}) - \lambda_{19}^P \varphi_{,11}^{(0)} + \lambda_{20}^P \theta_{,1}^{(0)} \\
&\quad + \lambda_{18}^P (u_{3,22}^{(0)} + u_{2,2}^{(1)}) - \lambda_{19}^P \varphi_{,22}^{(0)} + \lambda_{20}^P \theta_{,2}^{(0)} \\
&\quad + \lambda_{30}^P (n^{(0)} - N_0), \\
f_7^k &= -\lambda_{21}^P \varphi_{,11}^{(1)} + \lambda_{22}^P \theta_{,1}^{(1)} - \lambda_{21}^P \varphi_{,22}^{(1)} + \lambda_{22}^P \theta_{,2}^{(1)} \\
&\quad - [\lambda_3^P (u_{1,1}^{(0)} + 0.5 u_{3,1}^{(0)} u_{3,11}^{(0)}) + \lambda_8^P (u_{2,2}^{(0)} + 0.5 u_{3,2}^{(0)} u_{3,22}^{(0)}) \\
&\quad - \lambda_{23}^P \varphi_{,11}^{(1)} + \lambda_{20}^P \theta_{,1}^{(0)}], \\
f_8^k &= -\lambda_{24}^P n_{,1}^{(0)} \varphi_{,1}^{(0)} - \lambda_{24}^P n_{,11}^{(0)} \varphi_{,11}^{(0)} - \lambda_{25}^P n_{,1}^{(1)} \varphi_{,1}^{(1)} \\
&\quad - \lambda_{25}^P n_{,11}^{(1)} \varphi_{,11}^{(1)} + \lambda_{26}^P n_{,11}^{(0)} - \lambda_{24}^P n_{,2}^{(0)} \varphi_{,2}^{(0)} - \lambda_{24}^P n_{,22}^{(0)} \varphi_{,22}^{(0)} \\
&\quad - \lambda_{25}^P n_{,2}^{(1)} \varphi_{,2}^{(1)} - \lambda_{25}^P n_{,22}^{(1)} \varphi_{,22}^{(1)} + \lambda_{26}^P n_{,22}^{(0)}, \\
f_9^k &= -\lambda_{25}^P n_{,1}^{(0)} \varphi_{,1}^{(0)} - \lambda_{25}^P n_{,11}^{(0)} \varphi_{,11}^{(0)} - \lambda_{25}^P n_{,1}^{(1)} \varphi_{,1}^{(1)} \\
&\quad - \lambda_{25}^P n_{,11}^{(1)} \varphi_{,11}^{(1)} + \lambda_{27}^P n_{,11}^{(1)} - \lambda_{25}^P n_{,2}^{(0)} \varphi_{,2}^{(0)} - \lambda_{25}^P n_{,22}^{(0)} \varphi_{,22}^{(0)} \\
&\quad - \lambda_{25}^P n_{,2}^{(1)} \varphi_{,2}^{(1)} - \lambda_{25}^P n_{,22}^{(1)} \varphi_{,22}^{(1)} + \lambda_{27}^P n_{,22}^{(1)} \\
&\quad - \lambda_{24}^P n_{,1}^{(1)} \varphi_{,1}^{(1)} + \lambda_{26}^P n_{,1}^{(1)},
\end{aligned} \tag{36}$$

$$\begin{aligned}
f_{10}^k &= -\lambda_{28}^P \theta_{,11}^{(0)} - \lambda_{28}^P \theta_{,22}^{(0)} - s^{(0)}, \\
f_{11}^k &= -\lambda_{29}^P \theta_{,11}^{(1)} - \lambda_{29}^P \theta_{,22}^{(1)} + \lambda_{28}^P \theta_{,11}^{(0)}.
\end{aligned} \tag{37}$$

In this formulation, the same underlying physical model is used for each subtask to solve the nonlinear multi-field coupling problem of PS plate-like structures. The accuracy of the prediction results is evaluated using a loss function. For the k th subtask, the loss function is defined in the same manner as eq. (1), see eq. (B.6) given in Appendix B for details. To ensure smooth transitions between adjacent subtasks, the data points at their interfaces satisfy the conditions specified in eq. (14). The overall loss function for the complete model is given in eq. (4). The plate theory-based DD-PINNs-TD network architecture, designed specifically for addressing nonlinear multi-field coupling problems in plate-like PS structures, is provided in Figure A.2 in Appendix A. Moreover, the training and prediction processes using the DD-PINNs-TD model are illustrated in Figure 1(a).

4 Nonlinear multi-field coupling analysis of PS structures via DD-PINNs-TD models

Using the developed 3D theory-based, beam theory-based, and plate theory-based DD-PINNs-TD models in the previous section, we examine the nonlinear multi-field coupling mechanical responses of PS structures under different mechanical loadings and combination actions of mechanical and thermal loadings in this section. As numerical examples, we consider an n-type ZnO beam and plate with an initial electron concentration of $N_0=10^{21} \text{ m}^{-3}$. The beam-like structure (see Figure 3(b)) has dimensions: $a=20 \text{ mm}$, $b=6 \text{ mm}$, and $c=4 \text{ mm}$, and the plate-like structure (see Figure 3(c)) has dimensions: $a=8 \text{ mm}$, $b=4 \text{ mm}$, and $c=0.4 \text{ mm}$. For the 3D PS structure, the beam-like structure and the plate-like structure, and the generalized “clamped” boundary conditions are considered. Namely, there are: $u_i(x_1=0/a, x_2, x_3)=0$, $\varphi(x_1=0/a, x_2, x_3)=0$, $n(x_1=0/a, x_2, x_3)=10^{21} \text{ m}^{-3}$, and $\theta(x_1=0/a, x_2, x_3)=0$ for the 3D PS structure; $u_a^{(0)}(x_1=0/a)=0$, $u_1^{(1)}(x_1=0/a)=0$, $\varphi^{(0)}(x_1=0/a)=0$, $\varphi^{(1)}(x_1=0/a)=0$, $n^{(1)}(x_1=0/a)=0$, $\theta^{(0)}(x_1=0/a)=0$, $\theta^{(1)}(x_1=0/a)=0$, and $n^{(0)}(x_1=0/a)=10^{21} \text{ m}^{-3}$ for the beam-like structure; $u_i^{(0)}(x_1=0/a, x_2)=0$, $u_a^{(1)}(x_1=0/a, x_2)=0$, $\varphi^{(0)}(x_1=0/a, x_2)=0$, $\varphi^{(1)}(x_1=0/a, x_2)=0$, $\theta^{(0)}(x_1=0/a, x_2)=0$, $\theta^{(1)}(x_1=0/a, x_2)=0$, $n^{(1)}(x_1=0/a, x_2)=0$, and $n^{(0)}(x_1=0/a, x_2)=10^{21} \text{ m}^{-3}$ for the plate-like structure. The material properties of the ZnO are from ref. [71], that is, elastic constant (10^{10} N/m^2): $c_{11}=c_{22}=20.97$, $c_{33}=21.09$, $c_{13}=c_{23}=10.51$, $c_{12}=12.11$, $c_{44}=c_{55}=4.247$, $c_{66}=4.43$; piezoelectric constant (C/m^2): $e_{31}=e_{32}=-0.573$, $e_{33}=1.32$, $e_{15}=e_{24}=-0.48$; dielectric constant (10^{-11} F/m): $\bar{\epsilon}_{11}=\bar{\epsilon}_{22}=8.1103$, $\bar{\epsilon}_{33}=12.017$; pyroelectric constant ($10^{-6} \text{ C/m}^2 \text{ K}$): $\bar{p}=-9.4$; thermal stress constant (10^5 N K/m^2):

$\lambda=6.3629$; electron mobility: $\mu_{ij}=0.02$ ($\text{m}^2/(\text{V s})$); electron diffusion constant ($10^{-4} \text{ m}^2/\text{s}$): $D_{ij}=5.1774$; heat conduction constant ($\text{W}/(\text{m K})$): $\kappa=29.8$.

4.1 Modeling of beam-like PS structures under mechanical/thermal loads

We use the developed 3D theory-based DD-PINNs-TD model and beam theory-based models to predict the multi-field coupling mechanical responses of beam-like PS structures under only mechanical loading, and combined mechanical and thermal loadings. The load modes have three ways, that is, Case 1: a uniform pressure $q_F=1$ MPa is applied on the upper surface of the PS beam; Case 2: different uniform pressures ($q_F=0.6, 0.8$, and 1 MPa) are applied on the upper surface of the PS beam; Case 3: when the fixed uniform pressure $q_F=0.6$ MPa, different temperature changes ($\theta=10, 20$, and 30 K) are applied on the beam's right end.

4.1.1 3D theory-based DD-PINNs-TD predictions

For Case 1, the 3D theory-based DD-PINNs-TD model used in this example consists of 5 hidden layers with 50 neurons per layer and is trained over 50000 training epochs. The training dataset contains 2000 points collected from the surface of the PS beam. Figure 4 shows the distributions of the extensional displacement u_1 , the extensional displacement

u_2 , the flexural displacement u_3 , the electric potential ϕ , and the electron concentration n for different activation functions. It can be seen that the results obtained using the 3D theory-based DD-PINNs-TD model (with $M=2$) are in good agreement with the COMSOL simulations when the Tanh activation function is employed. In the COMSOL model, the computational domain is discretized using 5760 hexahedral elements, with each element consisting of 20 nodes based on quadratic Lagrange interpolation. When the ReLU activation function is used, the distributions of physical quantities remain quantitatively consistent with the COMSOL results; however, the magnitude of the error is larger. In the case of the Sigmoid activation function, while the distributions of displacement u_2 , displacement u_3 , and electric potential ϕ are consistent with the COMSOL results, discrepancies are observed in the extensional displacement u_1 and electron concentration n . These discrepancies are likely caused by overfitting in the neural network, which can be improved by either enlarging the training dataset or implementing early stopping during the training process.

Figure 5 presents the distributions of displacement u_3 , electric potential ϕ , and electron concentration n along the AB line (as indicated in Figure 3(a)) for different activation functions. In particular, Figure 5(c) highlights a significant difference in the electron concentration distribution when comparing the ReLU activation function and the Sigmoid or

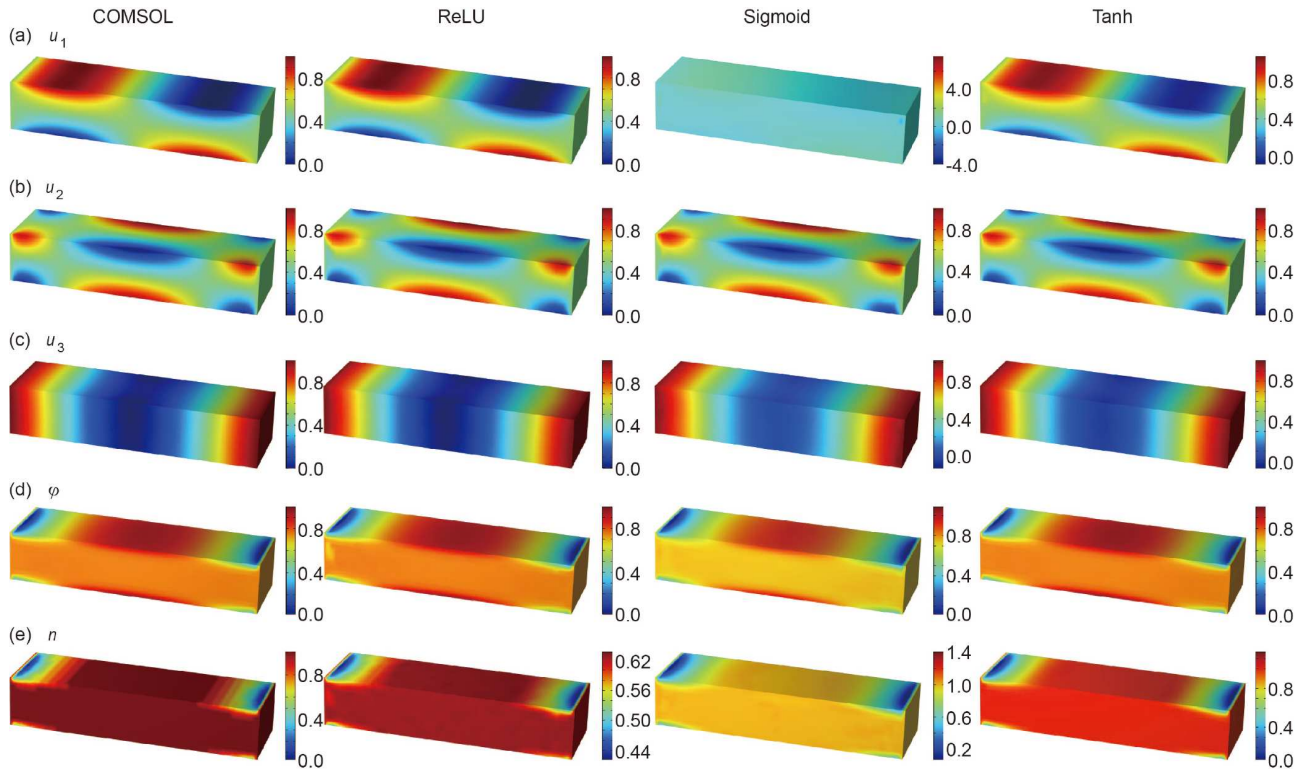


Figure 4 (Color online) The distributions of the (a) displacement u_1 , (b) displacement u_2 , (c) displacement u_3 , (d) electric potential ϕ , and (e) electron concentration n for different activation functions by the proposed 3D theory-based DD-PINNs-TD ($M=2$) model under $q_F=1$ MPa, and compared with COMSOL.

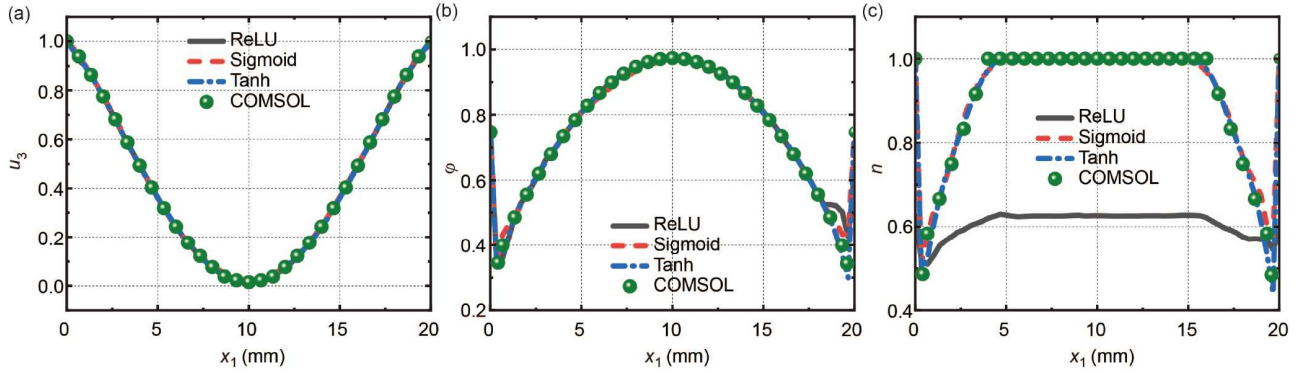


Figure 5 (Color online) The distributions of the (a) displacement u_3 , (b) electric potential ϕ , and (c) electron concentration n , for different activation functions at the line AB under $q_F=1$ MPa.

Tanh functions. This variation is attributable to the inherent physical nonlinearity incorporated in the developed nonlinear multi-field coupling model, notably the nonlinear drift current. Since ReLU is a piecewise linear activation function, it fails to accurately capture these nonlinear effects, thereby introducing large errors. In contrast, the Tanh active functions, being nonlinear, result in an electron concentration distribution that closely matches the COMSOL simulation results. Numerical results (Figures 4 and 5) further reveal the intrinsic characteristics of multi-field coupling mechanisms in PS structures, particularly the nonlinear complexity associated with carrier transport behavior. The distribution of electron concentration is not only influenced by external fields, but also predominantly governed by the internal nonlinear drift-diffusion mechanisms of the material.

Since the configuration of hidden layers and neurons plays a key role in the performance of the model. We now examine several examples to optimize the neural network training. Table 1 presents the MSE of the 3D theory-based DD-PINNs-TD model under different configurations of hidden layers and neurons and different levels of noise for various activation functions. It can be observed that when the ReLU activation function is used, the loss value is significantly higher compared to when the Sigmoid or Tanh functions are employed. In contrast, the Tanh activation function consistently yields both stable and low loss values. In fact, Table 1 shows that the smallest loss value is achieved when using the Tanh function with 7 hidden layers and 50 neurons. Consequently, for all subsequent examples, the neural network is configured with 7 hidden layers, 50 neurons per layer, and Tanh as the activation function. To further evaluate the robustness of the developed DD-PINNs-TD model, we assess its performance using training datasets with varying levels of disturbance. The evaluation results, listed in Table 1, reveal how the model responds to different disturbances, thus indicating its robustness or sensitivity to variations in the training dataset. Notably, for the Tanh activation function, the loss value remains consistently low when the dis-

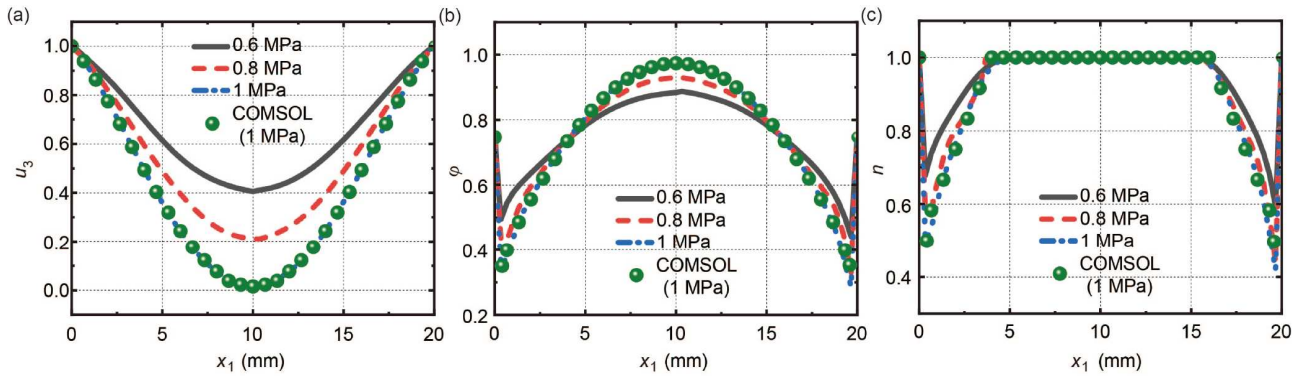
turbance level is below 5%.

For Case 2, we use the 3D theory-based DD-PINNs-TD model to predict the multi-field coupling responses of the beam-like PS structure under different uniform mechanical loads. During the numerical simulations, training datasets corresponding to various loading conditions are constructed. The DD-PINNs-TD model is then employed to predict the relevant physical quantities for each loading case, including displacement components and electric potential distributions. This approach allows the model to effectively capture the multi-field coupled responses of the structure under different external excitations. Figure 6 shows the distributions of the flexural displacement u_3 , electric potential ϕ , and electric concentration n as obtained from the 3D theory-based DD-PINNs-TD model (with $M=2$). It can be seen that these physical quantities increase with the magnitude of the applied uniform mechanical loads. Moreover, when comparing the results at $q_F=1$ MPa with those from the COMSOL simulation, the 3D theory-based DD-PINNs-TD model demonstrates excellent agreement, confirming its ability to accurately capture and predict the electromechanical behaviors of beam-like PS structures within the nonlinear multi-field coupling model. Figure 6 shows that all physical quantities increase with the applied mechanical load. The variation of electron concentration with loading reveals carrier migration behavior under the coupled effects of mechanical stress, electric field, and carrier dynamics, highlighting a pronounced nonlinear drift-diffusion mechanism.

Next, we evaluate the capability of the 3D theory-based DD-PINNs-TD model under complex combined mechanical and thermal loadings. For Case 3, we examine the multi-field coupling responses of the beam-like PS structure under different temperature changes (using the 3D theory-based DD-PINNs-TD model with $M=2$). Figure 7 shows the distributions of the extensional displacement u_1 , extensional displacement u_2 , flexural displacement u_3 , electric potential ϕ , electron concentration n , and temperature change θ along the line AB (see Figure 3 (a)) for different temperature changes.

Table 1 The MSE of the DD-PINNs-TD ($M=2$) method under different N (neurons) and L (hidden layers), and different levels of noise for different activation functions

Activation functions	Different N and L				Different levels of noise			
	$N=20, L=7$	$N=50, L=7$	$N=20, L=9$	$N=50, L=9$	0%	1%	5%	10%
ReLU	148.25	148.14	148.20	148.13	148.14	148.10	148.18	148.80
Sigmoid	3.8716	0.2768	28.589	0.6847	0.2768	0.4345	0.8365	3.1098
Tanh	0.4918	0.1770	0.6762	0.2761	0.1770	0.1637	0.7467	2.5405

**Figure 6** (Color online) The distributions of the (a) flexural displacement u_3 , (b) electric potential ϕ , and (c) electron concentration n by the proposed 3D theory-based DD-PINNs-TD ($M=2$) model under different mechanical loads at the line AB .

It can be seen that the extension displacement u_1 , extension displacement u_2 , flexural displacement u_3 , electric potential ϕ , and electron concentration n all increase with the increase of θ . When $\theta=30$ K, the predictions of the proposed 3D theory-based DD-PINNs-TD model are in very good agreement with the COMSOL simulations. Figure 7 shows that, under constant mechanical loading, all physical quantities, including displacement, electric potential, and electron concentration, increase with rising thermal loads. Due to the piezoelectric effect, external loads induce a polarization electric field, which further drives the directional mobility of electrons. The evolution of electron concentration with loading reveals a complex carrier transport process governed by the coupling of mechanical, electrical, thermal, and carrier-related fields, highlighting a pronounced nonlinear drift-diffusion mechanism. These coupling effects interact and evolve synergistically, representing the typical response characteristics of PS structures under complex loading conditions.

Figure 8 shows the distributions of the extensional displacement u_1 , flexural displacement u_3 , electric potential ϕ , electron concentration n , and temperature change θ under the mechanical load $q_F=0.6$ MPa and a temperature change $\theta=30$ K. Once again, the proposed 3D theory-based DD-PINNs-TD model shows excellent agreement with the COMSOL results, thereby demonstrating its accuracy and reliability in predicting the multi-field coupling mechanical behaviors of beam-like PS structures under complex combined mechanical and thermal loads. Overall, these results

demonstrate that the 3D theory-based DD-PINNs-TD model, when properly configured with an optimal network architecture and the Tanh activation function, can accurately predict the electromechanical behaviors of PS structures under a wide range of loading conditions.

4.1.2 Beam theory-based DD-PINNs-TD predictions

In this section, we employ the beam theory-based DD-PINNs-TD model to analyze the multi-field coupling mechanical behavior of the beam-like PS structure with the same dimensions as those presented in Figure 4. For Case 1, the neural network model is configured with 5 hidden layers, each containing 50 neurons. For the training dataset, we use the numerical results obtained from the established nonlinear beam model using COMSOL, where a total of 800 beam elements are used to discretize the boundary, with each element consisting of 3 nodes based on quadratic Lagrange interpolation functions; 2000 collocation points are randomly selected from these results to form the training dataset. The optimization is performed using the Adam optimization scheme over 20000 epochs.

Figure 9 shows the distributions of the extension deformation u_1 , flexural deformation u_3 , electric potential ϕ , and electron concentration n computed using the ReLU, Sigmoid, and Tanh activation functions. These results, obtained from the beam theory-based DD-PINNs-TD (with $M=4$) model, are compared with those from COMSOL simulations. As shown in Figure 9, the overall results from the proposed beam theory-based DD-PINNs-TD model align

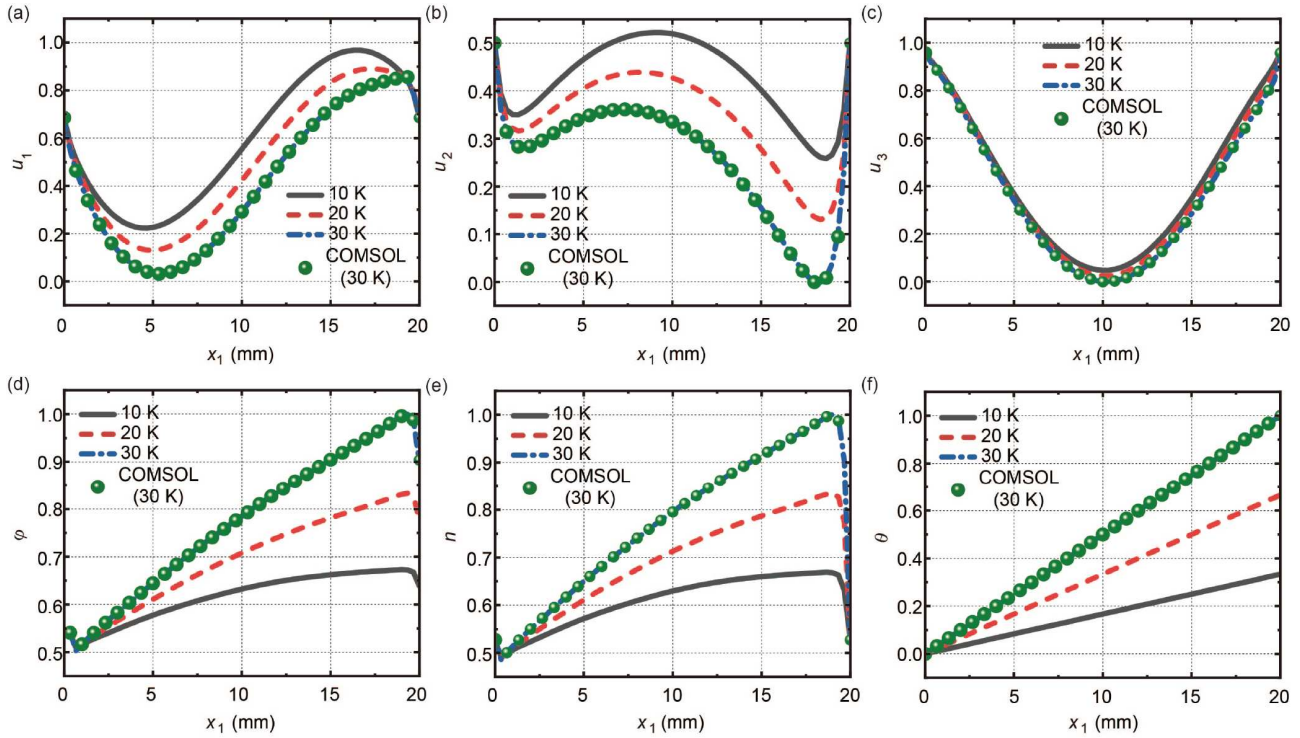


Figure 7 (Color online) The distributions of the (a) extensional displacement u_1 , (b) extensional displacement u_2 , (c) flexural displacement u_3 , (d) electric potential ϕ , (e) electron concentration n , and (f) temperature change θ by the proposed 3D theory-based DD-PINNs-TD model under different temperature changes at the line AB .

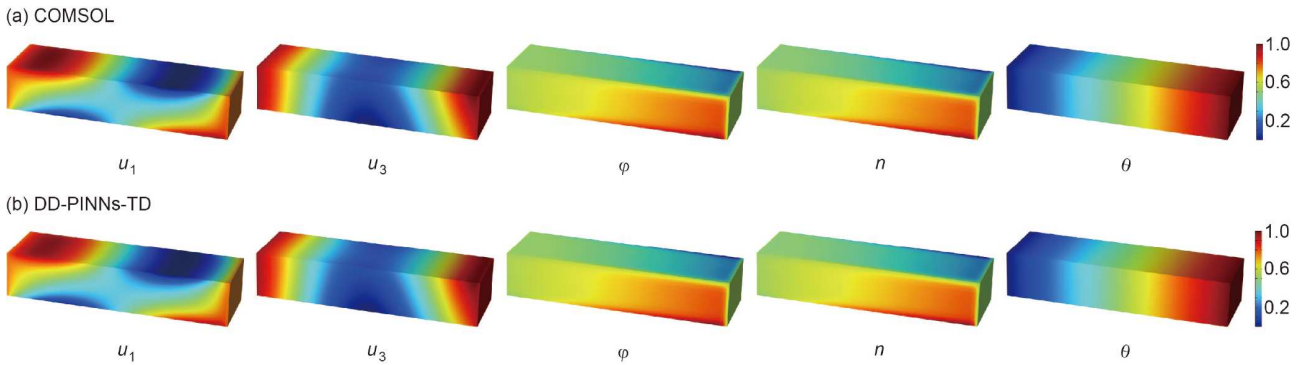


Figure 8 (Color online) The distributions of the extensional displacement u_1 , flexural displacement u_3 , electric potential ϕ , electron concentration n , and temperature change θ under $q_F=0.6$ MPa and $\theta=30$ K. (a) COMSOL; (b) 3D theory-based DD-PINNs-TD model.

closely with the COMSOL results, with satisfactory accuracy and low absolute errors for all field variables. However, it is observed that the absolute error is relatively higher near the clamped edges of the beam-like PS structure for the displacement fields, electric potential, and electron concentration. This is attributed to the approximate nature of the beam theory-based DD-PINNs-TD model at the boundaries where the displacement and electric conditions are strictly enforced.

Additionally, Figure 10 illustrates the distributions of the absolute error between the beam theory-based DD-PINNs-TD model and the nonlinear coupling model results from COMSOL for each field variable. The error distributions

indicate that, although there are larger deviations near the clamped edges, these deviations do not significantly affect the solution accuracy in the interior regions of the beam, as shown in Figure 10. Figure 11 shows the distributions of the displacement u_3 , electric potential ϕ , and electron concentration n along the line AB of the beam-like PS structure (in Figure 3(b)) as computed by the beam theory-based DD-PINNs-TD model using different activation functions, compared with the COMSOL simulations. It can be seen that the beam theory-based DD-PINNs-TD model obtained with different activation functions agrees well with the COMSOL simulations. Under uniform load conditions, the maximum

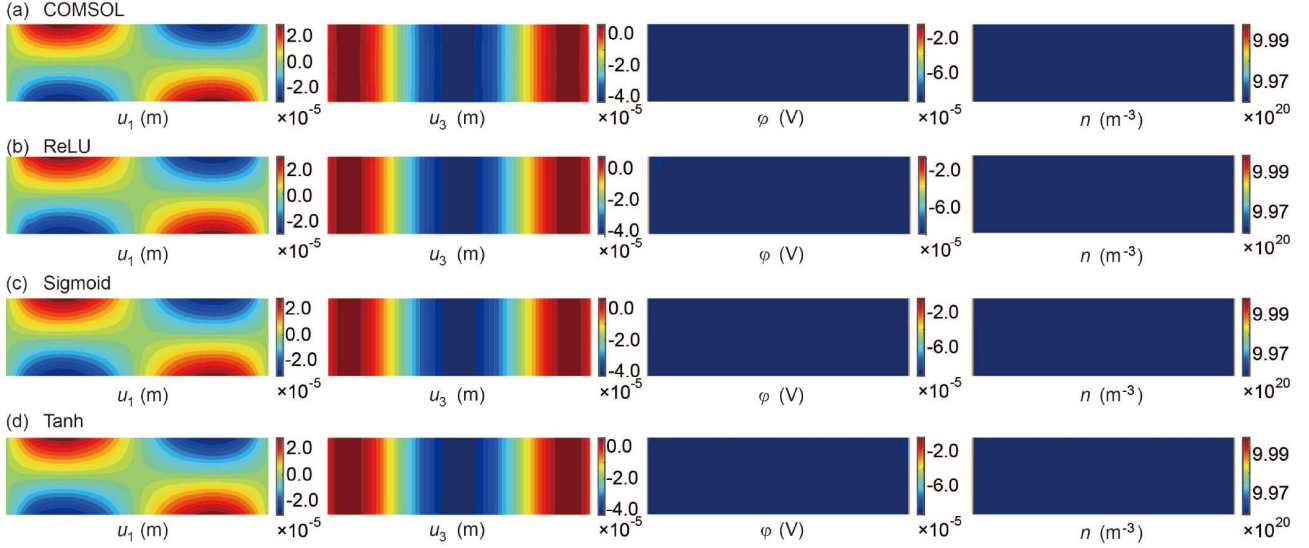


Figure 9 (Color online) The distributions of the extensional deformation u_1 , flexural deformation u_3 , electric potential ϕ , and electron concentration n of the PS beam-like structure by the (a) COMSOL and beam theory-based DD-PINNs-TD ($M=4$) model, (b) ReLU activation function, (c) Sigmoid activation function, and (d) Tanh activation function.

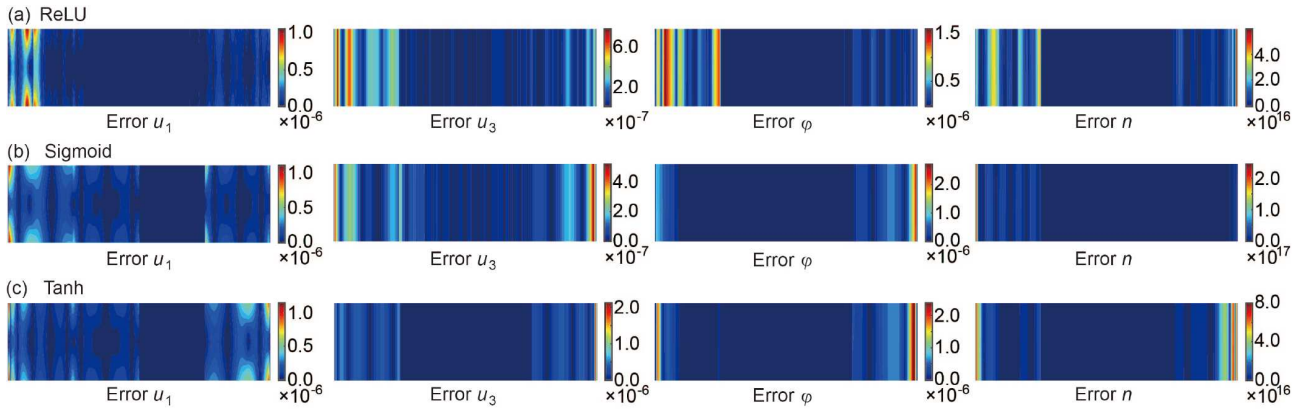


Figure 10 (Color online) The distributions of the absolute error between the COMSOL and beam theory-based DD-PINNs-TD model predictions associated with each field variable for the beam-like PS structure. (a) ReLU activation function; (b) Sigmoid activation function; (c) Tanh activation function.

electric potential is observed at both ends of the beam-like PS structure. This is attributed to the polarization electric field induced by mechanical load, which drives electrons toward the ends, as illustrated in Figure 11(c) and (d).

To select the most appropriate neural network configuration and activation function, we investigate the loss values for different numbers of neurons and hidden layers across different activation functions. Table 2 presents the MSE of the beam theory-based DD-PINNs-TD ($M=4$) model for different network configurations and activation functions. Analyzing the data in Table 2 alongside Figures 10 and 11 reveals that the Tanh activation function yields stable and accurate prediction distributions with relatively low loss values. Consequently, the subsequent neural network configuration employs the Tanh activation function. Notably, as shown in Table 2, the smallest loss value is achieved when

the network is configured with 20 neurons per layer and 7 hidden layers. This configuration will be used for the remainder of the examples.

Next, we investigate the influence of different mechanical loadings ($q_F=6 \times 10^5$, 8×10^5 , and 1×10^6 N/m) on the multi-field coupling mechanical responses of the beam-like structure using the beam theory-based DD-PINNs-TD (with $M=4$) model. Figure 12 shows the corresponding distributions of the flexural displacement u_3 , electric potential ϕ , and electric concentration n along the line AB (see Figure 3(b)). It can be seen that these physical quantities increase with the increase of the applied mechanical loads. For $q_F=1 \times 10^6$ N/m, the beam theory-based DD-PINNs-TD model predictions are in good agreement with the COMSOL results, demonstrating the beam theory-based DD-PINNs-TD model's ability to accurately capture the nonlinear multi-field coupling beha-

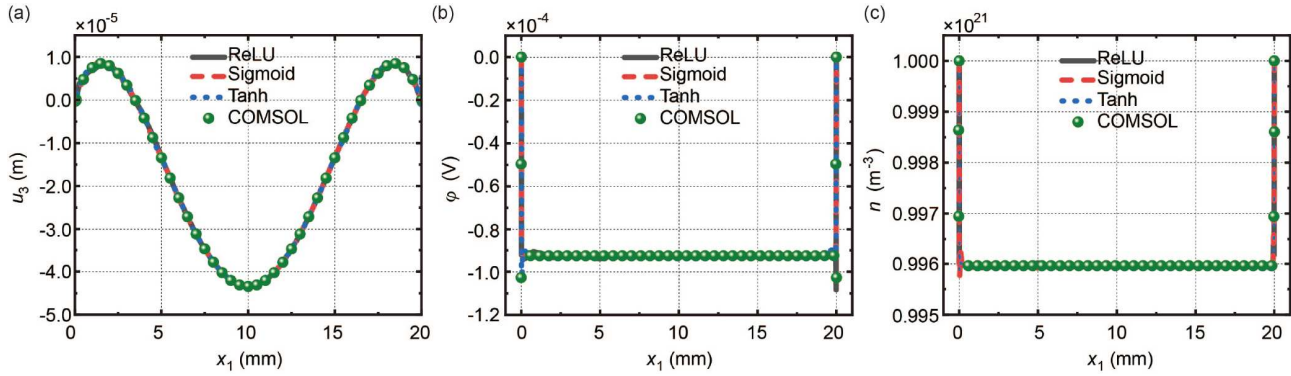


Figure 11 (Color online) The distributions of the (a) flexural deformation u_3 , (b) electric potential ϕ , and (c) electron concentration n via beam theory-based DD-PINNs-TD ($M=4$) model at the line AB of the beam-like PS structure under $q_F=1$ MPa.

Table 2 The MSE of the beam theory-based DD-PINNs-TD ($M=4$) model under different hidden layers and neurons for different activation functions

Layers and neurons	ReLU	Sigmoid	Tanh
$N=50, L=5$	0.1123	0.7667	0.4693
$N=20, L=7$	0.3192	0.6964	0.3500
$N=50, L=7$	0.4046	0.9011	0.6199
$N=20, L=9$	3.2360	0.6913	0.5609
$N=50, L=9$	0.6997	0.8748	0.6272

vior of beam-like PS structures.

Subsequently, for Case 3, we use the beam theory-based DD-PINNs-TD (with $M=4$) model to evaluate the flexural displacement u_3 , electric potential ϕ , and electron concentration n for temperature changes $\theta=10, 20$, and 30 K (with $\theta=0$ K as the reference temperature), and plot them in Figure 13. It can be seen from Figure 13(b) and (c) that as θ increases, both ϕ and n increase. Figure 13(c) shows that higher temperature changes cause electrons to gradually migrate from the center of the beam-like PS structure towards the edges. This behavior is driven by the temperature change-induced polarization electric field. In contrast, Figure 13(a) indicates that the influence of temperature change on the displacement fields is negligible. When compared with the COMSOL results (for $\theta=30$ K), the beam theory-based DD-PINNs-TD model shows high accuracy in predicting the physical quantities.

Figure 14 shows the distributions of the extensional displacement u_1 , flexural displacement u_3 , electric potential ϕ , and electron concentration n as computed by both the beam theory-based DD-PINNs-TD model and COMSOL simulation under the condition of $q_F=6 \times 10^5$ N/m and $\theta=30$ K. It can be observed that the proposed beam theory-based DD-PINNs-TD predictions for all these physical quantities are in good agreement with those obtained from the COMSOL model.

Through the analysis of the aforementioned work, we have

validated the effectiveness of the proposed 3D theory-based DD-PINNs-TD model and beam theory-based DD-PINNs-TD model for solving nonlinear multi-field coupling mechanical problems. Subsequently, we will further elucidate the advantages of the proposed method over traditional PINNs in terms of accuracy and computational time cost. The network configurations in Figures 4 and 9 are used in Tables 3 and 4, respectively. Tables 3 and 4 list the loss values and computation time cost for different activation functions via the 3D theory-based DD-PINNs-TD model and the beam theory-based DD-PINNs-TD model, respectively, compared to those achieved using traditional PINNs.

From Table 3, the results clearly show that the 3D theory-based DD-PINNs-TD model significantly outperforms the traditional PINNs model. Specifically, when using the Tanh activation function, the loss value is reduced by 58.97% and the computation time cost by 47.29%, relative to the traditional PINNs model. It is also noted that employing the ReLU activation function in the 3D nonlinear multi-field coupling model results in a higher loss value, since ReLU acts as a linear function and may not capture the inherent nonlinearities as effectively. Notably, in Table 4, the loss values and computation time costs are significantly lower for the beam theory-based DD-PINNs-TD model across various activation functions. Specifically, when $M=4$, the loss values achieved with ReLU and Sigmoid activation functions are reduced by over 70% and 90%, respectively. Additionally, the computation time using ReLU and Tanh activation functions is reduced by more than 50%. In comparison with Table 3, the use of the beam theory-based DD-PINNs-TD model dramatically decreases the computation time while maintaining accuracy. It is important to note that when working with smaller datasets, a significant portion of the computation time may be consumed by post-processing tasks (including prediction data handling and storage), which can increase the overall time cost. For further details on the computation time cost for $M=2$ and $M=4$ using ReLU and Sigmoid, please refer to Table 4. Unlike finite element

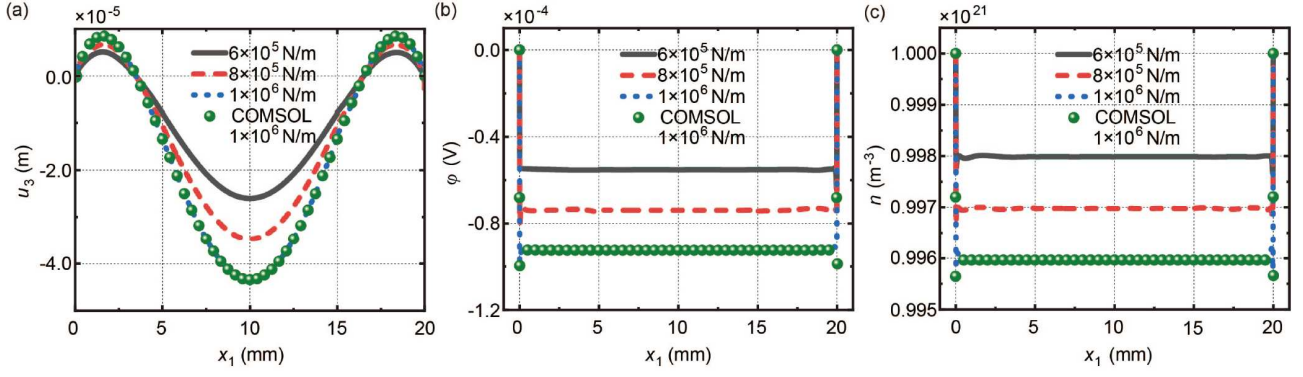


Figure 12 (Color online) The distributions of the (a) flexural deformation u_3 , (b) electric potential ϕ , and (c) electron concentration n via beam theory-based DD-PINNs-TD method under different mechanical loads at the line AB of the beam-like PS structure.

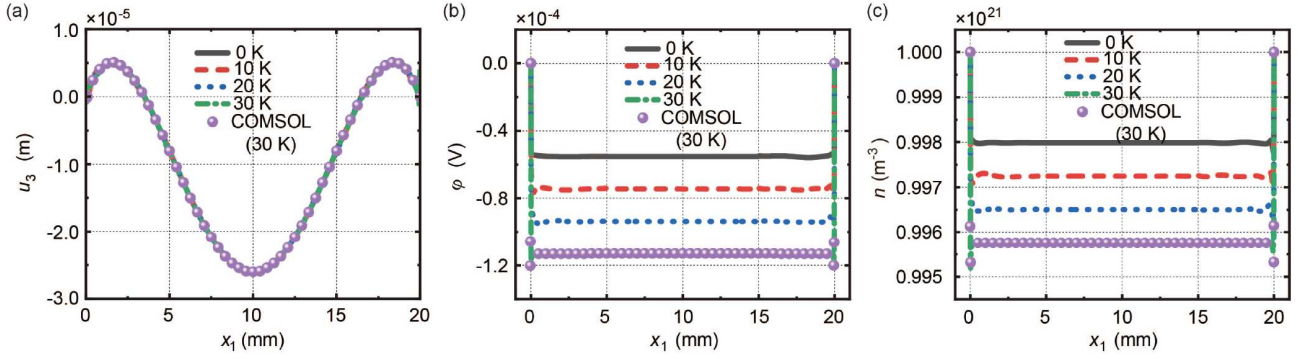


Figure 13 (Color online) The distributions of the (a) flexural deformation u_3 , (b) electric potential ϕ , and (c) electron concentration n via beam theory-based DD-PINNs-TD under different temperature changes at the line AB of the beam-like PS structure under $q_F=6 \times 10^5$ N/m.

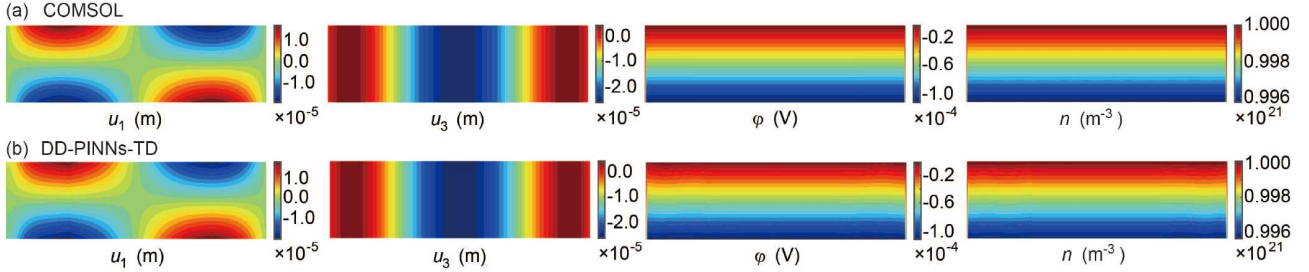


Figure 14 (Color online) The distributions of the extensional deformation u_1 , flexural deformation u_3 , electric potential ϕ , and electron concentration n of the beam-like PS structure under $q_F=6 \times 10^5$ N/m and $\theta=30$ K. (a) The results of COMSOL; (b) the beam theory-based DD-PINNs-TD ($M=4$) model.

meshing, selecting the appropriate number of divisions (M) in our approach should be based on specific requirements. A larger M is not always advantageous; increasing the number of divisions can result in fewer collocation points per sub-domain, which may lead to greater inaccuracies in the training model. Nonetheless, the proposed beam theory-based DD-PINNs-TD model offers significant advantages for handling large datasets and complex physical models. Overall, the results demonstrate that the proposed beam theory-based DD-PINNs-TD model effectively captures the nonlinear multi-field coupling behavior of beam-like PS structures, offering enhanced computational efficiency and

accuracy compared to traditional PINNs.

4.2 Modeling of PS plate-like structures under mechanical/thermal loads

In this subsection, we employ both the 3D theory-based DD-PINNs-TD model and the plate theory-based DD-PINNs-TD model to model the multi-field coupled mechanical behaviors of plate-like PS structures. By comparing them, we highlight the advantages of the plate theory-based DD-PINNs-TD model over the 3D theory-based model in terms of accuracy and computational cost. As numerical examples,

Table 3 The MSE of the PINNs and 3D theory-based DD-PINNs-TD model under different activation functions

Activation function	PINNs		3D theory-based DD-PINNs-TD ($M=2$)		
	Loss	CT (s)	Loss	CT (s)	AT (%)
ReLU	296.68	2873	148.15	1302	54.68
Sigmoid	0.6016	21037	0.3586	14760	29.84
Tanh	0.3822	22046	0.1568	11621	47.29

Table 4 The MSE of the PINNs and beam theory-based DD-PINNs-TD model under different activation functions

Activation function	PINNs		Beam theory-based DD-PINNs-TD					
	Loss	CT (s)	$M=2$			$M=4$		
			Loss	CT (s)	AT (%)	Loss	CT (s)	AT (%)
ReLU	1.6850	400.1	0.9352	175.5	56.1	0.1123	192.1	52.0
Sigmoid	3.0560	891.2	1.7171	594.3	33.3	0.7667	646.2	27.5
Tanh	0.8135	1446.2	1.1411	631.5	56.3	0.4693	592.7	59.0

both models use the same network configuration: the network is configured with 5 hidden layers, each containing 50 neurons, and the training process runs for 30000 epochs. The training dataset consists of 2000 sampling points.

4.2.1 3D theory-based DD-PINNs-TD predictions

First, as shown in Figure 3(c), a uniform pressure $q_F=0.6$ MPa is applied on the upper surface of the plate-like PS structure. Figure 15 shows the distributions of the extensional displacement u_1 , the extensional displacement u_2 , the flexural displacement u_3 , the electric potential ϕ , and the electron concentration n , computed by the 3D theory-based DD-PINNs-TD ($M=4$) model using different activation functions, and compared with the COMSOL results. Where the computational domain is discretized using 14400 hexahedral elements with quadratic Lagrange interpolation in the COMSOL model, and each element consists of 20 nodes. The results obtained using the 3D theory-based DD-PINNs-TD model with the Tanh activation function and ($M=4$) are in good agreement with COMSOL. Although the distributions obtained using ReLU and Sigmoid are qualitatively consistent with COMSOL, they exhibit larger errors.

Then, the 3D theory-based DD-PINNs-TD ($M=4$) model is used to evaluate the multi-field coupling responses of the plate-like PS structure under the combined action of a temperature change of $\theta=30$ K (applied to the right end) and uniform distributed mechanical load of $q_F=0.6$ MPa. The distributions of the flexural displacement u_3 , electric potential ϕ , and electron concentration n along the line AB (see Figure 3(c)) in Figure 16 and in the body are plotted in Figure 17, respectively. The 3D theory-based DD-PINNs-TD predictions under these combined loadings are in good agreement with COMSOL. In general, the predicted distributions of u_1 , u_2 , u_3 , ϕ , n , and θ are highly consistent with COMSOL,

demonstrating the method's accuracy in predicting various field variables for plate-like PS structures. Meanwhile, numerical results indicate that the polarization electric field induced by the combined mechanical and thermal loads drives electron migration, resulting in a redistribution of electron concentration. The variation in electron concentration reflects the carrier transport process under the synergistic influence of temperature, electric potential, and mechanical stress, revealing a typical thermo-electro-mechanical coupling mechanism. These multi-field coupling effects interact and evolve collaboratively, embodying the physical essence of the structural response.

4.2.2 Plate theory-based DD-PINNs-TD predictions

In this subsection, we employ the plate theory-based DD-PINNs-TD model to evaluate the nonlinear multi-field coupling responses of plate-like PS structures. Figure 18 presents the distributions of the extensional displacement $u_1^{(0)}$, extensional displacement $u_2^{(0)}$, flexural displacement $u_3^{(0)}$, electric potential $\phi^{(0)}$, and electron concentration $n^{(0)}$ as computed by COMSOL (The computational domain is discretized using 6000 quadratic quadrilateral elements, each consisting of 9 nodes based on Lagrange interpolation functions.) and the plate theory-based DD-PINNs-TD ($M=4$) model using ReLU, Sigmoid, and Tanh activation functions under $q_F=0.6$ MPa. Figure 19 further shows the absolute error distributions between the plate theory-based DD-PINNs-TD model and the COMSOL results for each field variable. The results in Figure 18 indicate that the comprehensive outcomes from the plate theory-based DD-PINNs-TD model using various activation functions closely match the COMSOL results, with minimal absolute errors across all fields.

We further use the plate theory-based DD-PINNs-TD

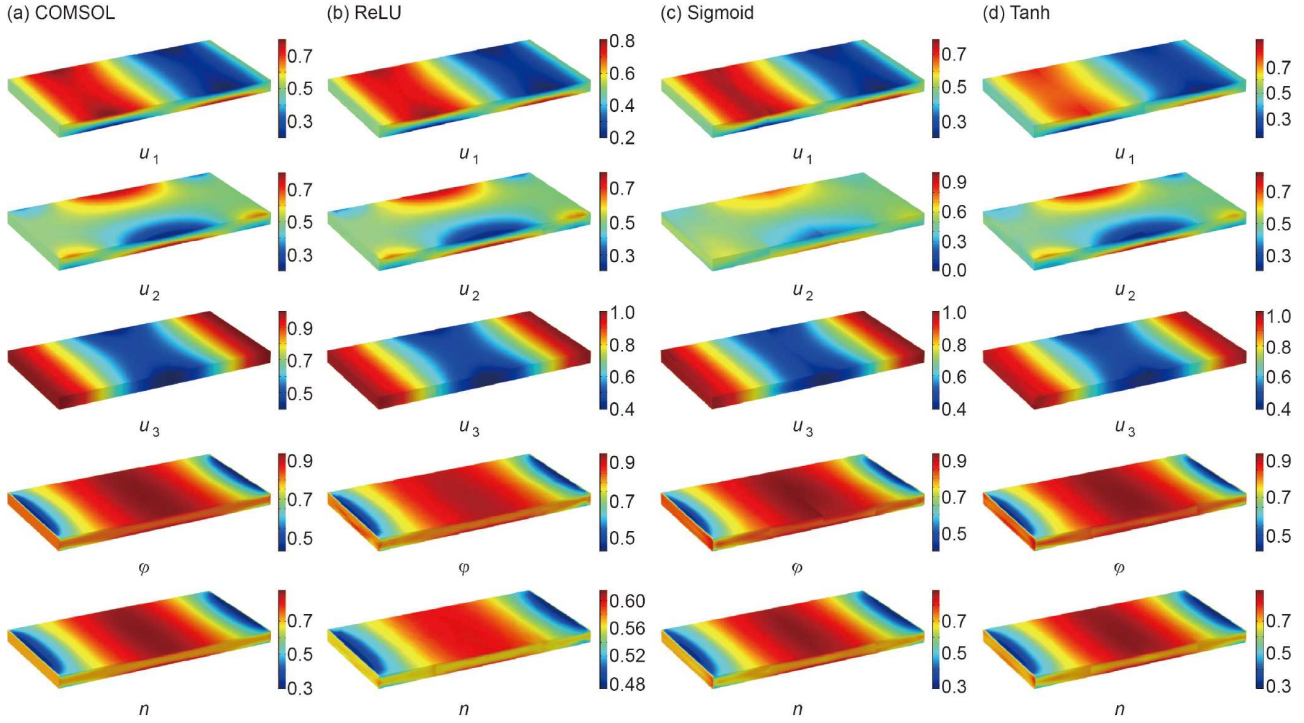


Figure 15 (Color online) The distributions of the displacement u_1 , the displacement u_2 , the displacement u_3 , the electric potential ϕ , and the electron concentration n under $q_F=0.6$ MPa for different activation functions. (a) COMSOL; (b) ReLU activation function; (c) Sigmoid activation function; (d) Tanh activation function.

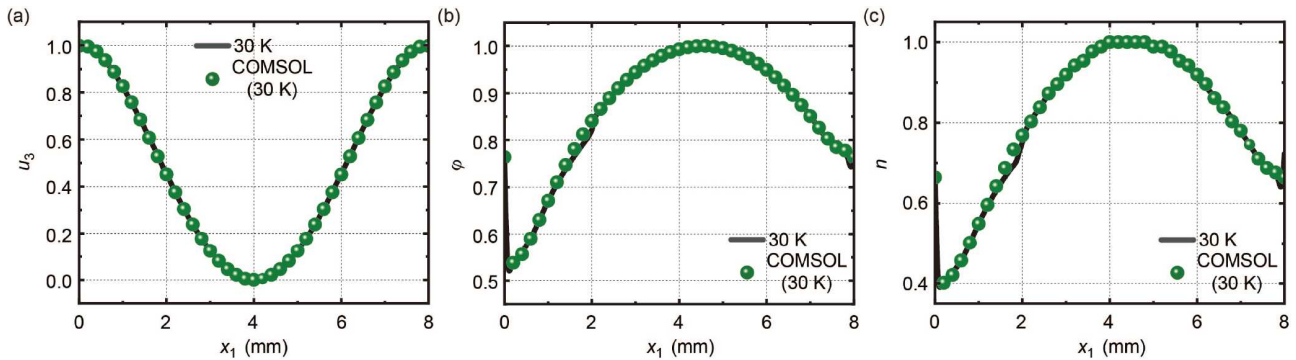


Figure 16 (Color online) The distributions of the (a) flexural displacement u_3 , (b) electric potential ϕ , and (c) electron concentration n by the 3D theory-based DD-PINNs-TD model at the line AB under $\theta=30$ K and $q_F=0.6$ MPa.

($M=4$) model to investigate the influence of different mechanical loads on the multi-field coupling mechanical behaviors of the plate-like PS structure under different uniform mechanical loads ($q_F=6\times 10^5$, 8×10^5 , and 1×10^6 N/m²). Figure 20 shows the distributions of the flexural displacement $u_3^{(0)}$, electric potential $\phi^{(0)}$, and electric concentration $n^{(0)}$ along the line AB (see Figure 3(c)). It can be seen that these physical quantities increase with the applied loads. The predictions under 1×10^6 N/m² are in good agreement with COMSOL, further demonstrating the method's reliability.

Finally, we examine the multi-field coupling responses of the plate-like PS structure under combined loading condi-

tions. In this case, fixed mechanical load $q_F=0.6$ MPa and temperature change $\theta=30$ K are applied on the upper surface and right end of the plate-like PS structure, respectively. Figure 21 shows the corresponding distributions of the flexural displacement $u_3^{(0)}$, electric potential $\phi^{(0)}$, and electric concentration $n^{(0)}$. Notably, Figure 21(c) shows that increasing θ causes electrons to gradually migrate from the left end toward the right end of the plate-like PS structure. This phenomenon is attributed to the temperature-induced polarization electric field. When compared with the COMSOL results at $\theta=30$ K, the plate theory-based DD-PINNs-TD model demonstrates excellent accuracy in predicting various

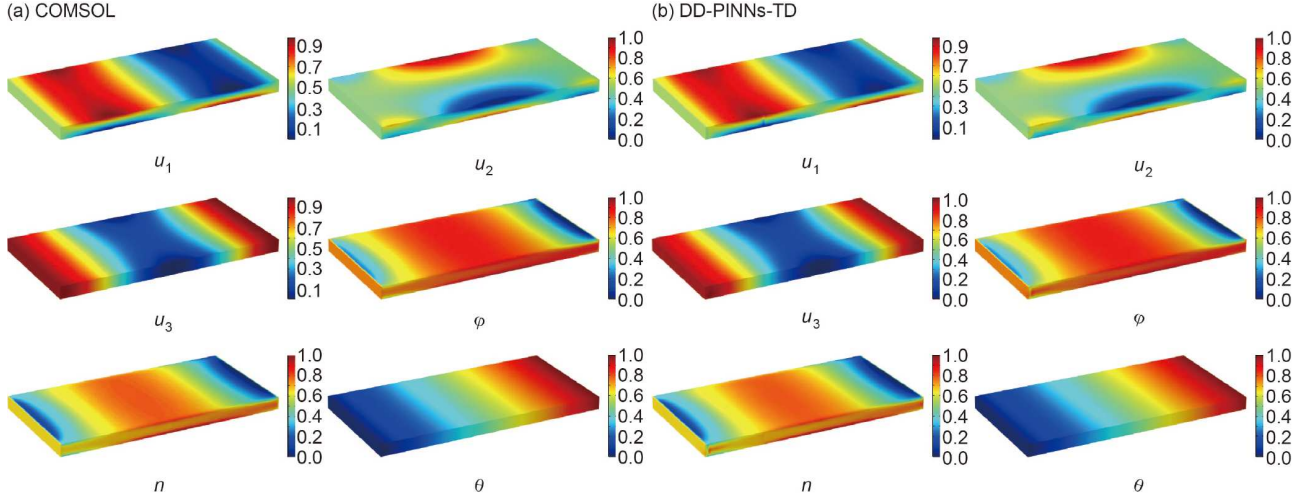


Figure 17 (Color online) The distributions of the displacement u_1 , the displacement u_2 , the displacement u_3 , electric potential φ , electron concentration n , and temperature change θ under $q_F=0.6$ MPa and $\theta=30$ K. (a) COMSOL; (b) the 3D theory-based DD-PINNs-TD model.

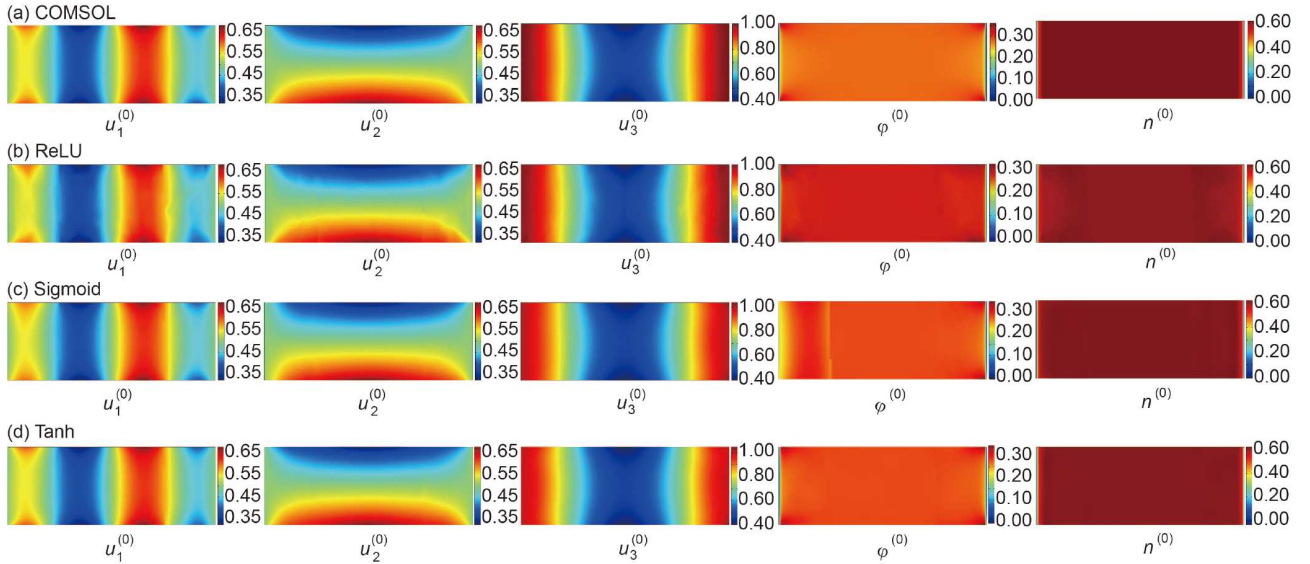


Figure 18 (Color online) The distributions of the extensional displacement $u_1^{(0)}$, the extensional displacement $u_2^{(0)}$, the flexural displacement $u_3^{(0)}$, extensional electric potential $\varphi^{(0)}$, and electron concentration $n^{(0)}$ by the (a) COMSOL and plate theory-based DD-PINNs-TD ($M=4$) model under $q_F=0.6$ MPa, (b) ReLU activation function, (c) Sigmoid activation function, and (d) Tanh activation function.

physical quantities. Figure 22 presents the distributions of the extensional displacement $u_1^{(0)}$, flexural displacement $u_3^{(0)}$, electric potential $\varphi^{(0)}$, electric concentration $n^{(0)}$, and temperature change $\theta^{(0)}$ as obtained from both COMSOL and the plate theory-based DD-PINNs-TD model under a load of $q_F=0.6$ MPa and $\theta=30$ K. The close agreement between the two sets of results further confirms that the proposed plate theory-based DD-PINNs-TD model effectively captures the nonlinear multi-field coupling mechanical behaviors of plate-like PS structures.

Next, we further elucidate the advantages of the proposed 3D theory-based DD-PINNs-TD and plate theory-based DD-

PINNs-TD models in terms of accuracy and computational cost compared to the traditional PINNs method. We use the same mesh configuration as shown in Figure 15. Tables 5 and 6 present the loss values and computational times of the 3D theory-based and plate theory-based DD-PINNs-TD models, respectively, compared with those of the conventional PINNs method under various activation functions.

As shown in Table 5, the proposed plate theory-based DD-PINNs-TD model significantly outperforms traditional PINNs in terms of both loss value and computation time. Notably, with the Tanh activation function (and with $M=4$ divisions), the loss value is reduced by 83.6% and the computation time is reduced by 82.5%. In contrast, when

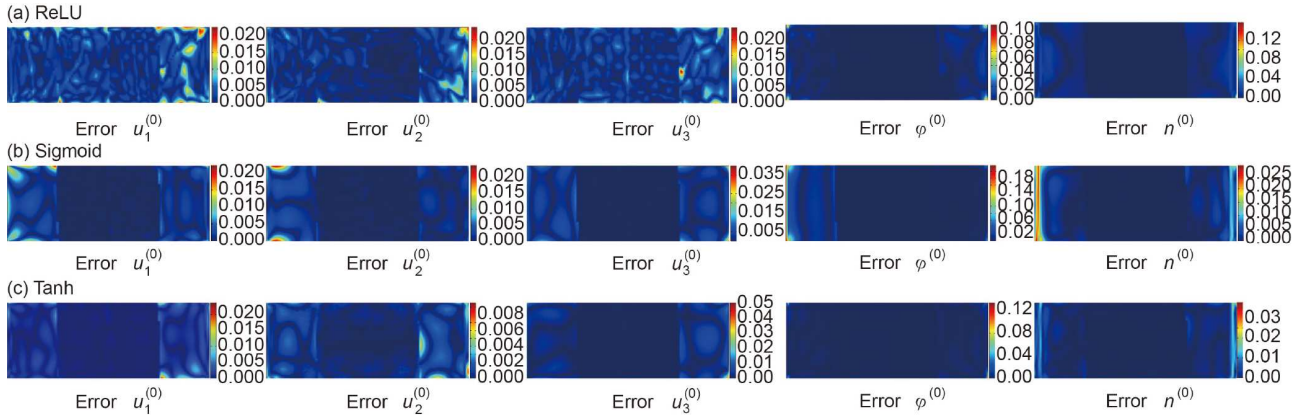


Figure 19 (Color online) The distributions of absolute error between the COMSOL and plate theory-based DD-PINNs-TD ($M=4$) predictions associated with each field variable for the plate-like PS structure. (a) ReLU activation function; (b) Sigmoid activation function; (c) Tanh activation function.

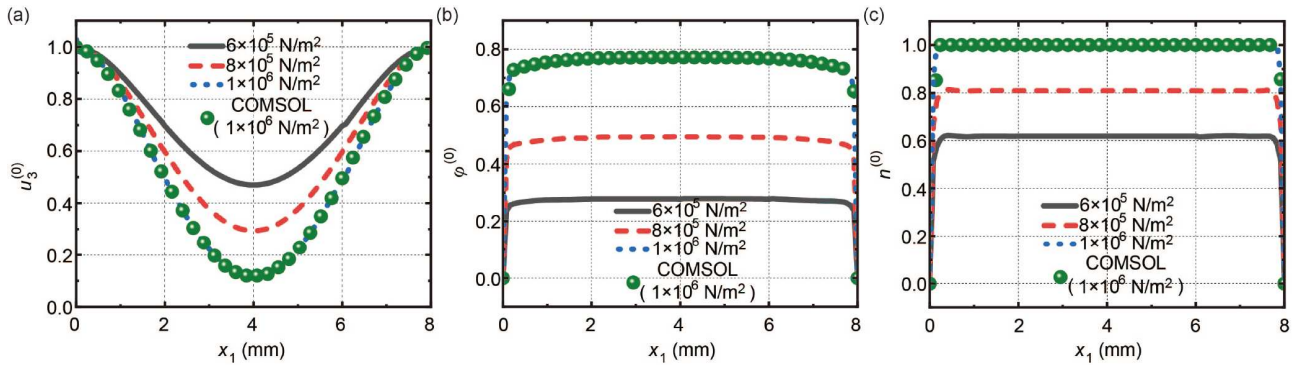


Figure 20 (Color online) The distributions of the (a) flexural displacement $u_3^{(0)}$, (b) electric potential $\phi^{(0)}$, and (c) electric concentration $n^{(0)}$ by the plate theory-based DD-PINNs-TD ($M=4$) model for different mechanical loads at the line AB .

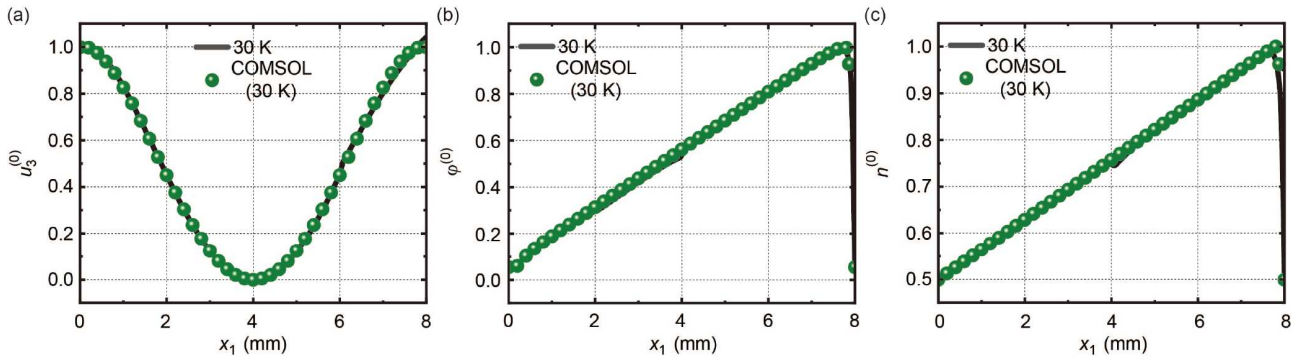


Figure 21 (Color online) The distributions of the (a) flexural displacement $u_3^{(0)}$, (b) electric potential $\phi^{(0)}$, and (c) electric concentration $n^{(0)}$ by the plate theory-based DD-PINNs-TD model at the line AB under $\theta=30$ K and $q_f=0.6$ MPa.

using the ReLU activation function, the loss remains relatively high, as discussed in Section 4.1.1 (see Table 1). Table 6 shows that the loss values and computation time for the plate theory-based DD-PINNs-TD model are significantly lower than those for traditional PINNs. Specifically, for $M=4$, the loss values with ReLU and Sigmoid are reduced by over 98% and 45%, respectively; and the computation time is reduced by more than 50% for ReLU, Sigmoid, and Tanh, with the Sigmoid activation function achieving a reduction

of over 80% in computation time. Compared with the 3D theory-based model (Table 5), the use of the plate theory-based DD-PINNs-TD model significantly improves both accuracy and computational efficiency.

5 Conclusion

In summary, the main contributions and findings of this

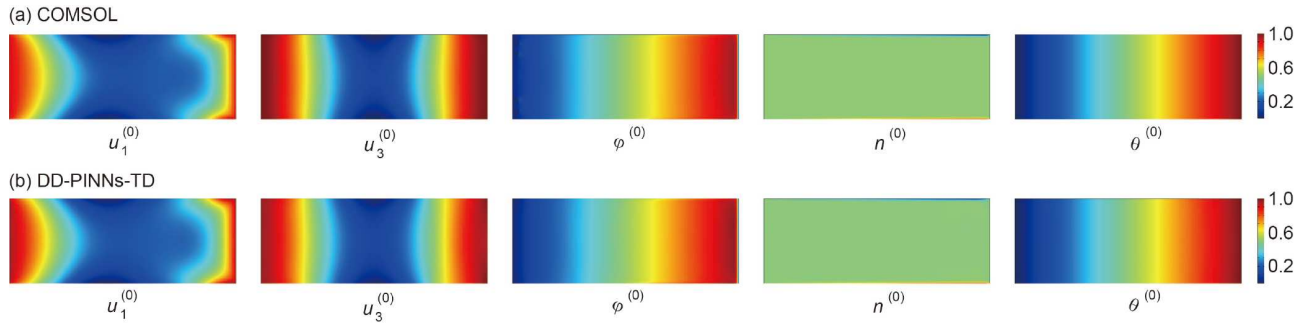


Figure 22 (Color online) The distributions of the extensional displacement $u_1^{(0)}$, flexural displacement $u_3^{(0)}$, electric potential $\varphi^{(0)}$, electric concentration $n^{(0)}$, and temperature change $\theta^{(0)}$ predicted by the (a) COMSOL, and (b) plate theory-based DD-PINNs-TD model under $q_F=0.6$ MPa and $\theta=30$ K.

Table 5 The MSE of the PINNs and 3D theory-based DD-PINNs-TD model for different activation functions for plate-like PS structures

Activation function	PINNs		3D theory-based DD-PINNs-TD					
	Loss	CT (s)	$M=2$			$M=4$		
			Loss	CT (s)	AT (%)	Loss	CT (s)	AT (%)
ReLU	92.171	2505	46.825	925	63.1	23.151	748	70.1
Sigmoid	0.7292	36028	0.7181	10745	70.2	0.2486	8404	76.7
Tanh	0.5578	32948	0.1385	8768	73.4	0.0914	5759	82.5

Table 6 The MSE of the PINNs and plate theory-based DD-PINNs-TD model under different activation functions for plate-like PS structures

Activation function	PINNs		Plate theory-based DD-PINNs-TD					
	Loss	CT (s)	$M=2$			$M=4$		
			Loss	CT (s)	AT (%)	Loss	CT (s)	AT (%)
ReLU	8.1694	1353	1.8045	783	42.1	0.1268	629	53.5
Sigmoid	0.4953	25664	0.1504	10759	58.1	0.2707	4746	81.5
Tanh	0.1440	19864	0.1386	6393	67.8	0.0306	4124	79.2

study are:

(1) A data-driven physics-informed neural networks (PINNs) via task-decomposition (DD-PINNs-TD) is proposed to address nonlinear multi-field coupling in material structures.

(2) The model demonstrates strong flexibility and robustness, especially under unknown material parameters.

(3) Numerical results show that DD-PINNs-TD improves prediction accuracy and reduces computational time by over 50% compared to traditional PINNs.

(4) The method effectively handles complex 3D problems under various loadings, and integration with structural theory further enhances computational efficiency and boundary accuracy.

The method proposed in this paper primarily addresses nonlinear multi-field coupling issues and is versatile, applicable to problems involving multi-material structures and multi-scale analyses. In the future, this approach could be integrated with shared neural networks by embedding shared parameters between sub-tasks, thereby enabling more flex-

ible handling of complex physical problems in predicting the mechanical behavior of material structures, ultimately achieving more accurate results, especially near the boundaries. Another avenue for development involves establishing an inverse design model for material structures, based on both the method presented in this paper and structural mechanics theory. Both applications will significantly contribute to the analysis of nonlinear problems in solid mechanics.

This work was supported by the National Natural Science Foundation of China (Grant Nos. 12172326 and 12192210), the Natural Science Foundation of Zhejiang Province (Grant No. LZ25A020007), and the National Key Research and Development Program of China (Grant No. 2020YFA0711700).

Conflict of interest The authors declare that they have no conflict of interest.

Supporting Information

The supporting information is available online at <http://phys.scichina.com> and <https://link.springer.com>. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific

accuracy and content remains entirely with the authors.

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